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REGULAR CONTRIBUTION

Advancements in memory-based cryptographic physical unclonable functions: a comprehensive review and future directions

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Abstract

With lightweight hardware security becoming increasingly critical for Internet of Things devices, Physical Unclonable Functions (PUFs) have emerged as a key enabling technology, providing device authentication, cryptographic key generation, and simplified key management capabilities without requiring dedicated on-chip key storage. While conventional MemPUFs have been extensively studied, they suffer from inherent limitations such as poor non-volatility and low entropy density. In recent years, emerging non-volatile memory PUFs built on Resistive Random Access Memory, Magnetoresistive Random Access Memory, Ferroelectric Random Access Memory, and Phase-Change Random Access Memory have shown tremendous potential. However, there remains a lack of systematic reviews and performance trade-off analyses that cover cutting-edge directions and provide practical guidance for real-world design. To this end, this paper systematically reviews the full technical lineage of MemPUFs and describes different emerging MemPUFs from two dimensions: entropy sources and development trends (including reliability, security, and energy efficiency). By statistically analyzing the performance metrics of representative works on emerging MemPUFs over the past five years, this paper reveals the critical impact of device selection on the overall performance of PUFs. On this basis, it systematically dissects the dual impact of Multi-Level Cell technology on PUF performance and prospectively discusses the "computation-security integration" architecture based on the deep convergence of Computing-in-memory and PUFs. The findings of this paper fill the gap in systematic reviews of emerging NVM PUFs and provide a valuable reference for the design of next generation low-power and high-security hardware security primitives for edge computing.

Keywords Cryptographic · Key management · Physical unclonable functions · Emerging non-volatile memory

1 Introduction

The rapid development of technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), and Deep Learning (DL) is transforming the technological landscape. According to the latest forecast from IoT Analytics, the global number of connected IoT devices is projected to reach 39 billion by 2030 and surpass 50 billion by 2035 [1]. This explosive growth in the mobile industry and the proliferation of interconnected devices underscore an urgent demand for robust security and privacy mechanisms in information storage. However, most

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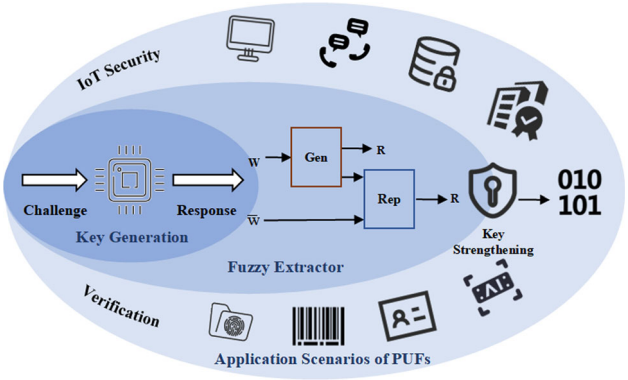


Fig. 1 Application scenarios of PUFs

encryption techniques rely on storing keys in volatile or non-volatile memory, such as SRAM or flash memory, which introduces potential vulnerabilities [2].

Malicious attackers pose a significant threat by cloning devices, extracting cryptographic keys, and replicating them to compromise users' private data. Data breaches are prevalent across various industries, creating substantial obstacles to the advancement of the cryptographic industry. This growing challenge highlights the urgent need for innovative hardware security primitives to safeguard the networked systems.

Physical Unclonable Functions (PUFs) represent an innovative security paradigm that exploits intrinsic physical variations in semiconductor devices to create cryptographic primitives. The foundational concept originated with Pappu et al. [3], who introduced the term Physical One-Way Function, later refined by Gassend et al. [4] through their formulation of the PUF terminology. This approach fundamentally differs from conventional cryptographic systems by eliminating the need for persistent key storage. Instead, PUFs generate cryptographic keys on demand through physical characteristics and immediately erase them after use, thereby establishing a new paradigm in secure hardware design. As shown in Fig. 1, native responses of the device require processing through fuzzy extractors to derive stable keys [5].

PUFs can be integrated with various schemes to enhance the security. For instance, PUFs can replace long-term stored static keys to improve secret sharing schemes [6], or be combined with TRNGs to strengthen masking schemes [7, 8]. As a hardware security primitive, PUFs demonstrate extensive applicability in IoT security architectures, particularly in key management [9–11] and cryptographic protocols [12–14].

Even among chips produced using the same manufacturing process, subtle differences in the physical hardware exist. These inherent variations enable PUF technology to generate a unique, unclonable "fingerprint", which plays a crucial role in preventing device counterfeiting and cloning. Depending on their operating principles, PUFs can be classified into delay-based and memory-based types. Memory-based PUF (MemPUFs), which leverage existing semiconductor process technologies such as SRAM and DRAM, do not require extensive modifications to the manufacturing process. This allows them to be implemented using mature technologies, thereby reducing development risks and manufacturing complexity. The significance of MemPUFs has gained increasing attention. Unfortunately, there is a relative lack of comprehensive review studies on the subject.

Regarding recent advances in review literature on PUFs, Gao et al. comprehensively reviewed the developmental trajectory of PUFs [15], explicitly demonstrating that raw responses require post-processing through a key generator before being deployed as cryptographic keys. Mahalat et al. [16] investigated novel PUF architectures designed for IoT environments with resource constraints. Existing survey literature predominantly focuses on PUF applications in IoT security protocols [17] and implementing PUFs using Field Programmable Gate Array (FPGA) [18], systematic investigations into the taxonomic characteristics of intrinsic PUF architectures and their operational mechanisms remains absent. Current reviews on silicon PUFs primarily concentrate on classification methodologies and Ring Oscillator PUFs [19], with limited in-depth discussion on architectural innovations employing emerging NVM technologies.

To address this research gap, this paper presents a systematic classification of PUFs based on their underlying physical mechanisms and operational principles, covering four major categories: strong PUFs, weak PUFs, delay PUFs, and MemPUFs. Unlike previous reviews that primarily focused on conventional silicon-based PUFs, this work also addresses the critical limitations of existing emerging MemPUF reviews, namely the lack of a unified evaluation framework and insufficient in-depth analysis of cutting-edge directions. The core contributions of this paper are as follows:

1. A comprehensive survey of MemPUFs covering the full technical lineage. We thoroughly review the technological evolution roadmap of MemPUFs from SRAM, DRAM, Flash to RRAM, MRAM, FRAM, and PRAM, and deeply analyze the entropy sources and implementation architectures of various emerging MemPUFs. On this basis, we systematically summarize the optimization pathways and technical bottlenecks of emerging MemPUFs over the past decade across three core dimensions: reliability, security, and energy efficiency.

2. A unified quantitative evaluation framework based on technology node normalization. We collect and statistically analyze the core performance parameters of representative works on various MemPUFs published between 2020 and 2026, and conduct a cross-sectional quantitative comparison of their energy efficiency, area overhead, reliability, and attack resistance. We quantitatively reveal the supe-

Table 1 Comparison between strong PUFs and weak PUFs

Characteristic	Strong PUF	Weak PUF
Number of CRPs	Exponential quantity	Linear quantity
Security	High	Low
Reliability	Moderate	High
Design Complexity	Complex	Simple

rior energy efficiency of hafnium-based ferroelectric devices compared to other emerging memory technologies.

3. A forward looking proposal for the paradigm of next generation MemPUFs. We systematically dissect the dual impact of Multi-Level Cell (MLC) technology on PUF performance, clarifying the new challenges introduced while enhancing entropy density and attack resistance. Furthermore, we deeply explore the core technical bottlenecks and potential solutions for the deep convergence of Computing-in-Memory (CIM) and PUFs, providing novel insights into the computation-security co-design for next-generation edge computing devices.

2 Classification of PUFs

There exist multiple classification criteria for PUFs. Among the widely accepted standards, two primary categories are recognized. First, based on their response mechanism, PUFs can be classified into Strong PUFs and Weak PUFs. Second, according to their physical implementation basis, PUFs are categorized into Electronic PUFs and Non-electronic PUFs.

2.1 Strong PUFs and weak PUFs

The Challenge-Response model is a standard operational framework for PUFs. When a challenge is presented to a PUF chip, it generates a corresponding response, creating a device-specific mapping called a Challenge-Response Pair (CRP). These CRPs are securely stored in embedded hardware devices to ensure confidentiality. PUFs are categorized by their available CRPs: Strong PUFs provide an exponentially large number of CRPs, while weak PUFs only support a small, finite set [21].

As shown in Table 1, weak PUFs generate a limited and controllable set of responses with lower sensitivity to environmental variations. Their responses are inherently more repeatable and stable, making them well-suited for applications such as generating fixed keys or device identifiers [22]. In particular, weak PUFs are ideal for scenarios involving one-time key generation, where the key exists only while the chip is powered on, making them especially appropriate for resource-constrained devices. However, the limited

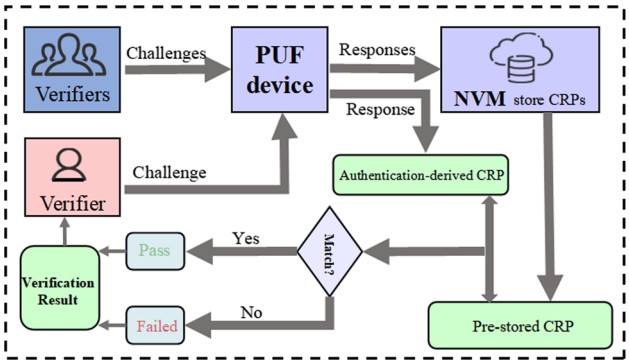


Fig. 2 The authentication process of PUFs. Verifiers refer to multiple registered users used to construct the database, while Verifier denotes a single user performing authentication

number of responses makes weak PUFs potentially easier to be compromised if an attacker can obtain or infer these CRPs. Consequently, they are less secure and unsuitable for applications requiring frequent interactions or complex authentication protocols.

Strong PUFs, in contrast, offer an exponential number of CRPs, providing greater flexibility and enabling their use in complex cryptographic protocols and scenarios requiring multiple authentications. Examples include identity authentication [23], and dynamic key generation. A typical PUF authentication flow is illustrated in Fig. 2. During the enrollment phase, the authenticator generates and sends a large number of challenges to the PUFs embedded in target devices, collecting and securely storing the corresponding CRPs in NVM for future use. During subsequent authentication requests, the system retrieves the stored CRPs to verify the device's identity. If the provided response matches the stored CRPs, authentication succeeds, granting the device access or execution privileges. If the response does not match, authentication fails, and the system rejects the request. It is worth noting that, despite their advantages, strong PUFs are more vulnerable to simulation attacks, such as machine learning (ML) modeling, due to the abundance of available CRPs. As such, ensuring response stability and optimizing against these vulnerabilities is critical. However, strong PUFs provide robust defense against certain types of attacks. Even if an attacker gains access to some CRPs, predicting new responses remains challenging, and reversing or cloning the PUF is exceedingly difficult.

2.2 Non-electronic PUFs and electronic PUFs

The physical properties of devices at the microscopic level are inherently random and cannot be precisely controlled during the manufacturing process. This randomness arises from material inhomogeneities, process variations, and minor defects introduced during fabrication. Based on the material

and the physical properties utilized, PUFs can be categorized into non-electronic PUFs and electronic PUFs. We present this classification in the Fig. 3, while other subcategories will be introduced in subsequent sections.

3 Non-electronic PUFs

Non-electronic PUFs are widely used in applications such as optical labeling and banknote anti-counterfeiting. Primarily including optical PUFs, paper PUFs, and CD PUFs. Optical PUFs leverage the unique optical behaviors of randomly distributed glass spheres embedded within a transparent disk. By directing a laser onto the disk, the system exploits laser incidence, scattering, and refraction to produce a unique response, achieving physical unclonability. In this setup, the excitations are the angles and positions of laser incidence, while the system response is the feature vector generated from light patterns processed by a computer. Paper PUFs [24] utilize the reinforcing fiber structure of paper as a unique identifier. When UV light scans the paper, the resulting image serves as the system's unique response. Similarly, early CD manufacturing processes introduced random imperfections, such as pits, scratches, and particles, due to material and process variations. During CD reading, these imperfections induce distinctive optical behaviors that are detected by photodetectors, generating unique signal patterns. These patterns can serve as identifiers, forming what is known as CD PUFs [25].

3.1 Electronic PUFs

Electronic PUFs can be divided into analog circuit PUFs, delay PUFs, and MemPUFs.

The coated PUF, introduced by Tuyls et al. [26], is one of the most representative PUFs based on analog circuits. This design utilizes a coating containing randomly distributed dielectric particles sprayed onto the substrate material. When a specific excitation is applied to the coating, local sensors embedded in the top metal layer of the integrated circuit generate capacitance values as the response. In the same year, the team enhanced the design of the coated PUF by improving the capacitance structure and proposed a related theoretical model [27]. Based on this work, they later introduced the LC-PUF [28], which is similar to the coated PUF. However, there is a significant limitation of these approaches. These designs often occupy a large area and require high power consumption, which is a critical drawback in modern IC applications that demand high integration and low power consumption. Subsequently, researchers began to investigate the stochastic behavior in nanoscale transistors. Oxide traps exhibit randomly distributed charge emission characteristics, where the emission time constant is gate voltage dependent and remains stable even after device aging. Leveraging this property, the designed PUF quantifies the emission probability of traps, thereby optimizing power consumption while significantly expanding the CRP space. This enhancement effectively mitigates ML attacks [29].

Delay PUFs primarily include Arbiter PUFs (APUFs), Ring Oscillator PUFs (RO PUFs), and Glitch PUFs. These PUFs generate unique responses directly from the physical characteristics of the circuit, require no additional NVM to store keys. This feature reduces power consumption and facilitates lightweight designs, enabling the integration with other types of PUFs. Typically implemented on FPGAs, these designs rely on the core logic unit, known as the Look-Up Table (LUT), to achieve picosecond-level, low-cost, and high-precision configurable delay lines. These delay lines are used to generate unique response signals [30]. By precisely configuring the LUT logic, delay paths can be fine-tuned to ensure the uniqueness and reliability of PUF responses under varying operating conditions.

APUFs. APUFs are based on the time delays inherent in interconnect lines and transistor gates caused by random variations in the manufacturing process [31]. APUFs utilize a linear adder structure [32], where two parallel signals compete as they traverse a chain of switching selectors to produce binary outputs [4]. The delay of an n stage path depends on the cumulative delays from the interconnects and selector chain at each stage. The delay difference at each stage is determined by the corresponding challenge signal. This structure supports additional input signals, facilitating the generation of more CRPs. However, response errors caused by unbalanced routing [33] reduce the reliability and uniqueness of APUFs. To address these issues, subsequent research efforts have introduced alternative designs such as k-XOR-APUFs [34, 35] and FF-APUFs[36].

RO PUFs. Similar to APUFs, the RO PUF [37] employs head-to-tail inverters, to form two signal transmission paths that oscillate at distinct frequencies. The PUF leverages small manufacturing variations in the inverters and interconnects to produce a stochastically unique response. However, RO PUFs require a lot of hardware resources during implementation [38], and environmental noise, such as fluctuations in voltage and temperature, thereby negatively impacting measurement stability and consistency [39].

Glitch PUFs. Conventional delay PUFs are susceptible to modeling attacks, as the relationship between their delay and generation information can be easily and accurately predicted. To address this vulnerability, Suzuki et al. proposed the delay structure of Glitch PUFs in 2010 [40]. Glitch PUFs leverage the phenomenon of competitive signal propagation within gate circuits to generate burr waveforms with nonlinear characteristics [41]. By precisely controlling circuit delay variations, Glitch PUFs produce "burr" signals during signal

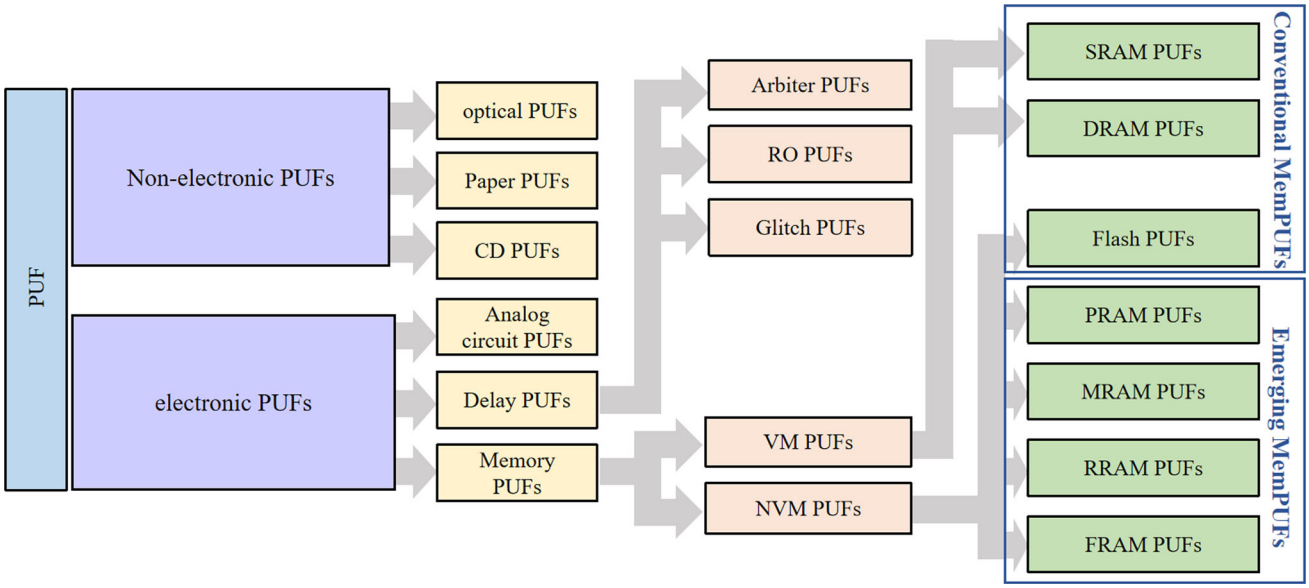


Fig. 3 PUFs can be classified into electronic PUFs and non-electronic PUFs based on their underlying materials and operational principles, each of which can be further subdivided into various specific PUF types

transmission. The waveform of these signals is influenced by differences in the delay lines within the gate circuits and variations in the transition times required for logical high-to-low and low-to-high transitions [42]. To generate a unique delay response, the burr signals are processed through waveform sampling, jitter correction, and related operations. This process converts the burr information into an output signal that reflects the intrinsic physical differences of the device.

It is crucial to consider that any delay in generating PUF responses could create a vulnerability, allowing attackers to exploit this timing gap to extract private information. Longer delays, particularly when the PUF design is computationally intensive, increase the likelihood of successful modeling attacks with high accuracy. With the development of semiconductor technology, researchers have begun exploring the inherent randomness introduced during the device manufacturing process. Among various architectures, MemPUFs directly leverage the physical characteristics of memory cells, offering superior advantages in terms of stability, area efficiency, power consumption and scalability.

4 Memory PUFs

Conventional memory technologies rely on well established charge storage mechanisms and CMOS based architectures, offering high reliability and scalability but facing limitations in power efficiency and endurance. In contrast, emerging non-volatile memory technologies, including RRAM, MRAM, and ferroelectric memory, leverage novel physical mechanisms such as resistance switching and spin polarization, enabling superior characteristics such as non-volatility, faster switching speeds, and higher energy efficiency. Consequently, MemPUFs can be categorized into those utilizing conventional technologies and those exploiting emerging technologies.

Figure 4 illustrates the unified workflow for feature extraction and generation in MemPUFs, which aims to reveal the underlying mechanism by which physical randomness evolves into a unique digital fingerprint. Entropy Source is the origin of PUF randomness. During chip fabrication, unclonable physical mismatches at nanoscale arise from uncontrollable process variations, including doping fluctuations, geometric mismatches, material nonuniformity, and other nanoscale disorders. Device-to-Device (D2D) variation provides the fundamental fingerprint, ensuring that different devices remain independent even after reconfiguration. Cycle-to-Cycle (C2C) variation offers the driving force for reconfiguration, enabling the device to generate new responses across different time or operating cycles. Memory cells are triggered by specific external stimuli, such as power-up signals, voltage pulses, or timing control sequences. These excitations drive the cells into a metastable state, initiating the competition of internal physical characteristics. State latching occurs through a dynamic competition process. Under the applied stimulus, the analog physical quantities within the cell compete with each other. Ultimately, the microscopic mismatch determines whether the cell settles into “State 0” or “State 1”. Cells with large mismatches latch quickly and stably, whereas those with weak mismatches exhibit probabilistic fluctuations due to noise. Finally, a read circuit, such as a sense amplifier or comparator, converts the latched ana-

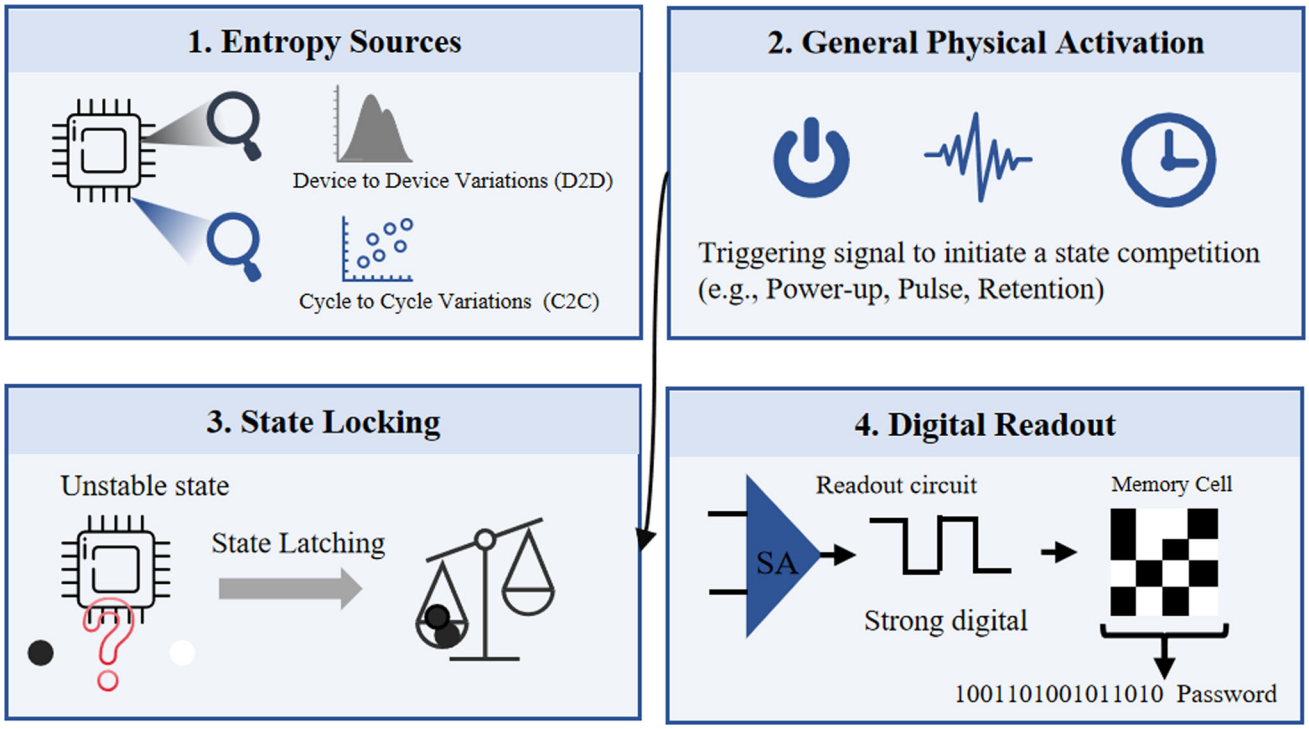


Fig. 4 Unified workflow for feature extraction and generation in MemPUFs

log physical state directly into a binary sequence known as the Raw Response.

4.1 Conventional MemPUFs

Conventional memory types such as SRAM and DRAM can be readily configured within existing chip architectures, enabling their deployment across diverse devices to safeguard information exchange in large-scale machine environments.

4.2 SRAM PUFs

The memory cell of an SRAM consists of two cross-coupled inverters, each capable of producing a unique random 1 or 0 state. Due to manufacturing variations, SRAM cells exhibit slight differences when powered on, making them suitable for use as a unique identifier of the chip. Nevertheless, irreversible device aging progressively modifies circuit characteristics, threatening the long-term reliability of authentication systems. Based on restrict race code, an aging resistant method for PUF authentication is realized [43]. Over the past two decades, numerous SRAM PUF designs have been proposed [44–48]. These advancements have led to several commercial solutions, such as $Quiddi Key400$, and $Securyzr^{TM}$. These solutions support the principles of root key endogeneity and private key non-storage ensuring the protection of sensitive data on the chip. One notable advantage of SRAM PUFs is their inherent "delete after use" capability for PUF derived keys, ensuring volatile key storage. As SRAM is a volatile memory, it cannot retain data after a power failure, ensuring that sensitive information is automatically erased once the chip is powered down. Research on MemPUFs initially focused on implementing PUFs using FPGAs. Intellectual Property (IP) protection has been a critical area of interest for FPGA vendors, leading to the development of various bitstream-based encryption schemes [21, 49]. However, not all FPGAs are equipped with internal SRAM. To address this limitation, the butterfly PUF [50] was introduced, leveraging a structure similar to SRAM. It uses the FPGA's internal matrix to characterize its physical properties.

4.3 DRAM PUFs

DRAM is widely used in most systems, making DRAM-based PUF one of the most extensively researched types of MemPUFs. DRAM is typically configured in a 1T1C structure, as depicted in Fig. 5a, where multiple DRAM cells are organized in an array structure. The word line controls transistor conduction, enabling data read and write operations through bit line control when the transistors are active. It is important to note that DRAM uses capacitors to store data, which are prone to data loss during operation and therefore require periodic refreshing. Since transistors function as cell switches, the inherent random variations in the devices are

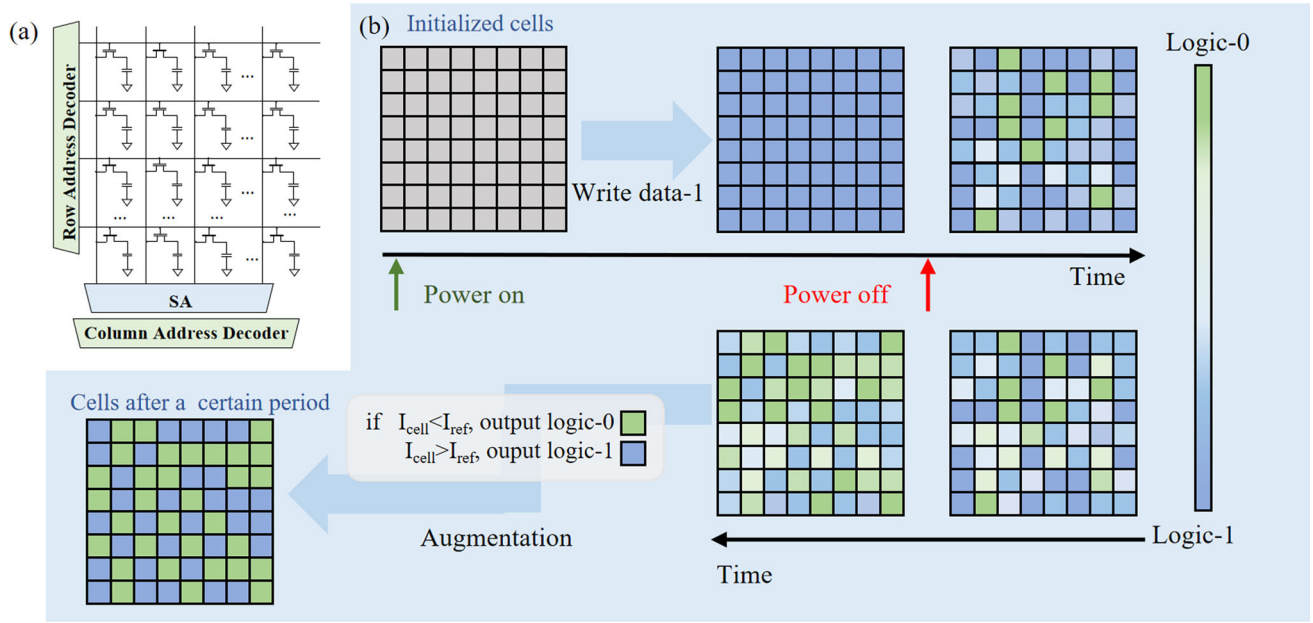


Fig. 5 **a** The 1T1C structure of DRAM. **b** After power-on, all DRAM cells are written to data "1". Following power-off, due to charge leakage, the cell current is read. If the I_{cell} is greater than the I_{ref}, it is pulled up to a high level and regarded as logic-1; otherwise, it is pulled down to a low level and regarded as logic-0

actually reflected in the charging and discharging characteristics of the capacitors. This phenomenon forms the basis for PUFs utilizing the up-charge value [51] and data residuals [52, 53]. The matrix in Fig. 5b provides an example, where blue cells represent logic-1 stored values and green cells represent logic-0. Initially, all DRAM cells are set to logic-1. After a period of data refresh cycles or power-off to accelerate charge leakage, some cells randomly transition to logic-0. Over a specific duration, the ratio of data-0 and data-1 stabilizes at approximately 50%. This randomness can be exploited to construct PUFs.

However, data residual-based PUFs are limited by the long decay cycles required for generating reliable responses. An alternative approach involves configuring the write cycle [54], which has been proven to be an effective method for generating PUF responses by adjusting the memory controller configuration. DRAM PUFs for cryptographic key generation have seen significant advancements, particularly in peripheral circuitry design. Recent work by Abulibdeh et al. [55] enhanced DRAM PUFs by introducing multiplexers at sense amplifier inputs and optimizing peripheral circuits. This design exploits bit-line propagation delay differences and extracts entropy through voltage comparisons between adjacent DRAM cells, resulting in a highly reliable PUF.

4.4 Flash PUFs

Logically, a NAND gate is a CMOS circuit, and NAND flash memory derives its name from the way its internal memory cells are connected, mimicking the logical function of a NAND gate. Lofstrom et al. pointed out that addressable MOSFET arrays produce random drain currents due to device mismatches, creating unique voltage sequences usable for ICID generation [56]. However, this approach was not considered secure due to its reliance on randomness generated by device mismatches, making it susceptible to environmental variations, aging effects. With the maturation of CMOS manufacturing processes, significant progress has been made in developing compact, low-overhead PUFs [57, 58]. Furthermore, even with deviations in device parameters caused by various factors, Bai et al. simulated circuit parameter variations resulting from fabrication process variability through Monte Carlo simulations [59], enabling the realization of a PUF from a simulation perspective.

In NAND flash memory, multiple storage cells are tightly packed to form a series structure. The interactions between these flash cells generate entropy, which can be leveraged for PUF applications [60, 61]. Due to the high storage density of NAND flash, it is possible to generate a large number of CRPs, which can be utilized for identification and authentication. The realization principles of this approach are as follows:

Enrollment Phase: During the enrollment phase of the PUF, one or more NAND pages are selected and erased to a known state. IDs of device are generated after programming these pages, using effects such as Programming Disturbance, Programming/Erase cycles, and Read Disturbance.

Verification Phase: The authentication process involves regenerating the device ID and performing a comparison with the pre-registered ID to verify legitimacy.

In the proposed PUF, the Fake Flash Detection comprehensive framework offers protection for devices in commercial off-the-shelf applications [62], while the extraction of specific flash chip family DNA [63] helps maintain the database during the PUF verification process.

Additionally, the physical limitations of planar NAND flash, such as challenges in miniaturization and electrical interference, have led to new directions in the development of 3D NAND. By stacking memory cells vertically, the storage density can be significantly increased. The presence of charge-trapping layers and the vertical stacking of storage cells depend heavily on the punch-through process. The non-uniform edge roughness of etched or photolithographically formed pattern edges can lead to parametric variations in storage cells, and inherent manufacturing process uncertainties further facilitate the development of PUFs in 3D NAND [64, 65].

4.5 Emerging MemPUFs

As memory technology approaches the limits of Moore's Law, conventional miniaturization becomes increasingly difficult. Emerging non-volatile memory technologies exhibit superior bit stability [66] while offering scalable capacity, low power consumption, and high reliability. These characteristics that make them particularly suitable for lightweight security implementations. These advantages have positioned such memory technologies as promising candidates for PUF applications, constituting a significant research focus in hardware security. Various emerging MemPUFs have been proposed, utilizing RRAM, MRAM, FRAM and PRAM [67]. Unlike conventional PUFs that rely on manufacturing process variations (e.g., SRAM PUFs) [68], emerging memory devices not only inherit randomness from process fluctuations but also introduce intrinsic randomness dominated by the novel materials themselves during operation, thereby forming a composite entropy source. This provides a promising capability for device authentication and secure key generation. But they are also plagued by new types of security vulnerabilities such as Row-hammer attacks and fault injection attacks [69].

4.6 RRAM PUFs

As an early-developed emerging MemPUF, RRAM PUF has been extensively and maturely investigated. We therefore first elaborate on the entropy source types of RRAM PUFs and the key challenges in their current development trends, including reliability, security, energy efficiency, and area overhead. Subsequently, we extend the discussion to research on other emerging MemPUFs, so as to provide a clear content overview and lay a solid foundation for the study of various emerging MemPUFs.

4.7 Entropy sources

The nonlinearity of current during RRAM programming and the random distribution of conductance can both be utilized as physical entropy sources for PUFs.

Liu et al. experimentally verified the feasibility of RRAM PUFs on a 1 kb HfO_2 1T1R RRAM array in 2015 [70]. Distinct current variations exist during the transition of RRAM from the low resistance state (LRS) to the high resistance state (HRS). In this scheme, cells with current above a reference current are defined as LRS to realize binary digitization. The reference current selection strategy significantly affects the uniqueness of the PUF: when the reference current is generated using only a single column of cells, the inter chip Hamming distance is merely 24.2%; when dummy cells from five arrays are used, the inter chip Hamming distance increases to 49.8%. Generally, employing more dummy cells for reference current generation improves the uniqueness [70]. Each memory cell in this design outputs one bit of response, classifying it as a weak PUF. Given that the HRS of RRAM exhibits a wider resistance distribution than the LRS [71], programming devices into the HRS yields approximately log-normally distributed resistance variations. Based on this characteristic, Chen et al. proposed a 1T1R array-based PUF architecture, in which the addresses of two n-bit data are applied as the challenge, and an n bit response is generated by comparing the resistance values of the selected two cells [72]. Subsequently, Pang et al. [73] implemented a 1kb RRAM PUF using the same scheme and demonstrated that the number of CRPs can approach 100 million in larger arrays, thus satisfying the requirements of strong PUFs. However, it should be noted that the CRP mapping in this architecture is approximately linear and has been proven vulnerable to ML attacks [74].

Different from the entropy sources mentioned above, Nili et al. generated random bits for PUF implementation by cumulatively comparing the linear currents of multiple passive RRAM devices [75, 76]. The key principle of this design is that memristors exhibit distinctive nonlinear I-V behavior, whose nonlinearity and corresponding variability become pronounced under relatively large bias voltages, thus serving

as a natural randomness source. By selecting and accumulating multiple columns for comparison, the number of CRPs can be significantly increased, enabling the construction of a strong PUF.

In addition, the random short term memory time can also be exploited as an entropy source for PUFs [77]. At present, the mainstream entropy sources for RRAM PUFs are mainly divided into two categories: one relies on the approximately normal distribution of resistance values, and the other utilizes the inherent nonlinear I-V characteristics of the devices. The resistance states of RRAM may drift after repeated read-write operations, resulting in unstable output responses. In contrast, schemes that directly employ the nonlinear I-V characteristics as the entropy source show superior stability and have become a research hotspot. It should be noted that this approach may introduce interference through sneak paths in crossbar arrays.

4.7.1 Development trends

Contemporary development trends of RRAM PUFs are gradually evolving toward high reliability, high security, low power consumption, and small area overhead.

4.7.2 Reliability issues

After comprehensively evaluating the key performance metrics of RRAM PUFs, Chen's research team identified reliability as the major challenge facing current RRAM PUFs [72, 78]. RRAM devices suffer from random telegraph noise, which introduces random fluctuations in the read current. Improvements in reliability require optimizations at the circuit level or the PUF architecture level.

(1) Optimizing readout circuits for enhanced reliability. These include adopting high sensitivity sense amplifiers [78], multiple sampling or extended sensing schemes [78], and improving calibration circuits [79].

(2) Improving PUF design strategies. Typical methods include multiple sampling and majority voting [78], as well as establishing a uniform initial state distribution prior to PUF response generation [73]. It should be noted that some memory cells may fail during device fabrication and operation, leading to bit errors. When resistance is used as the entropy source, multiple cells can be connected in parallel to generate the response [70]; when device current is used as the entropy source, the currents of multiple cells can be summed to produce the response [76]. This strategy not only improves reliability but also increases the sensing current, easing the design burden of the readout circuit.

In addition to intrinsic read current fluctuation and resistance drift, experimental studies have shown that RRAM PUFs are highly sensitive to temperature variation, parasitic capacitance, and transistor-level nonidealities RRAM structures, all of which can degrade reset probability and increase response instability [80].

4.7.3 Security issues

To resist ML modeling attacks, design strategies can be summarized into two categories: logic level and temporal level approaches.

In the logic dimension, the core idea of existing anti-ML schemes is to extend simple comparison operations into multi-level nonlinear transformations with digital mapping by introducing hidden challenges or auxiliary logic, thereby breaking the linear correlation between challenges and responses and significantly increasing mathematical complexity. A typical implementation is the two-layer RRAM PUF architecture [74, 81]. In this structure, the binary data generated by the first layer is mapped to the address of the second layer, and the output of the second layer serves as the final response. The challenge undergoes two rounds of array access and comparison, effectively breaking its linear relationship with the response. Another approach enhances nonlinearity by introducing mathematical operations. Mahmoodi et al. [76] generated an auxiliary bit (AUX) by comparing two fixed columns, while the main sense amplifier compared the remaining columns to generate a raw bit; the final response bit was obtained by XORing this raw bit with the AUX bit. This design eliminates output offset with extremely low overhead and further improves resilience against machine learning attacks.

In the temporal dimension, dynamic refresh of PUF fingerprints is realized using the physical reconfigurability of RRAM. Since the resistance distribution of cells varies dynamically across different switching cycles, reconfiguration can be easily implemented. C2C variation provides a natural foundation for constructing reconfigurable PUFs [82].

4.7.4 Energy efficiency and area issues

To improve energy efficiency, devices fabricated with CMOS compatible processes are adopted to simplify the circuit design of PUFs. Auxiliary sense amplifiers were hard-wired to the first and last columns to generate AUX bits, and the final response bit was obtained by XORing the main output bit with the AUX bit [76]. This design achieved an energy consumption of approximately 45 fJ per bit. Mahmoodi et al. pointed out that a smooth thin-film surface can be obtained by adjusting the etching process during device fabrication, thereby improving I–V uniformity. Meanwhile, a higher aspect ratio enables the integration of more sophisticated PUF designs without complex peripheral circuits during device fabrication [83], resulting in an energy consumption of only 4 fJ per bit.

Area optimization can be achieved through 3D stacking or optimized circuit structures. Yang et al. developed a vertical RRAM structure similar to 3D NAND, where stacking n layers reduces the area per output bit by a factor of n, significantly lowering area overhead [82]. Cao et al. noted that memory density is limited by selector size, and the use of shared selectors can substantially reduce the area of RRAM PUF arrays [79].

We summarize the two main implementations of RRAM PUF in Fig. 6, I_1 and I_2 can be the currents of any two selected cells, or the current of any single selected cell and a predefined reference current. When passive RRAM devices are used as the carrier for RRAM PUFs, they offer CMOS process compatibility and a small footprint. In PUF design, the readout circuit can be simplified in the back-end process to reduce power consumption, while for high-security designs, non-linear logic can be introduced in the back-end process to resist machine learning attacks. The entropy source of the PUF is typically derived from the inherent I–V characteristics of the devices. However, passive device arrays introduce additional sneak paths as interference, which enhances the randomness of the PUF. Furthermore, passive devices can be architecturally designed to enable 3D stacking, further reducing the area of the RRAM PUF. When active RRAM devices are employed as the carrier for RRAM PUFs, RRAM memory arrays are generally adopted, where the resistance states of RRAM cells are programmed to serve as the entropy source of the PUF. In such designs, the transistor gates can be precisely controlled such that only selected cells are conducting, eliminating sneak path interference, but at the cost of a larger area. The resistive characteristics of RRAM vary dynamically over different switching cycles. When an adversary may obtain full CRPs, the PUF can resist machine learning attacks by reconstructing the CRPs.

4.8 MRAM PUFs

With the advances in the field of spintronics [84, 85], spintronic devices are gradually becoming a research hotspot.

4.8.1 Entropy sources

An MRAM cell is based on a magnetic tunnel junction (MTJ), which consists of a ferromagnetic layer, a tunnel barrier, and another ferromagnetic layer. Typically, one ferromagnetic layer has a fixed magnetization direction and is called the reference layer, while the other can switch its magnetization and is referred to as the free layer. When the magnetic moments of the reference and free layers are parallel, the MTJ exhibits a low-resistance state; otherwise, it is in a high-resistance state. The intrinsic device variability of MRAM provides a natural entropy source for PUF implementations.

A typical MRAM PUF directly reads the high/low resistance states of multiple MTJ cells, with current passing directly through the MTJ. A reference resistor or adjacent cell is usually used for comparison to obtain a digital output. For example, Park et al. randomized the magnetization of the reference layer in a 70 nm MTJ array and used the ratio of anti-parallel resistance to parallel resistance (RAP/RP) as the output bit, yielding a direct binary output and eliminating the need for high-precision ADCs [86]. In addition, stochastic switching behavior and thermal fluctuations during read/write operations can further enhance randomness. When the energy barrier of the free layer is comparable to thermal energy, spontaneous spin flipping occurs (superparamagnetic effect), which can be exploited to apply random pulses to MTJs for generating random bits [87]. Moreover, the switching probability under write pulses and C2C variations also provide dynamic entropy sources [87, 88]. These physical differences result in unique resistance characteristics among memory cells, which can be used to generate device-specific responses. Similar to the operating mechanism of SRAM, the random bias of the free layer in MRAM after power-down or initialization can also be utilized. However, due to the relatively high energy barrier in MRAM, conventional spin-transfer torque MRAM (STT-MRAM) generally does not exhibit obvious random states upon power-off.

4.8.2 Development trends

In recent years, MRAM PUFs have shown a clear evolution from device-level randomness exploitation to system-level security optimization, similar to the trend of RRAM PUFs, mainly reflected in reliability enhancement, security improvement, and higher energy efficiency.

Reliability Issues

The co-design of sensing circuits and device states has become a key approach to improving the reliability. Existing studies have shown that stable and temperature-robust entropy sources can be achieved by regulating the magnetization distribution in the MTJ reference layer [86]. Furthermore, Chen et al. proposed a resistance state reconfiguration strategy to screen and reconstruct unstable cells, maintaining the bit error rate (BER) of MRAM PUF below 2.2% over a wide range of voltages and temperatures [89]. Similarly, Meng et al. reduced the worst-case BER from 0.146% to 0.0025% by introducing a low-offset sense amplifier, significantly improving read stability [90].

2. Security Issues

Studies using attack validation with models such as generative adversarial networks (GANs) have shown that high-quality MRAM entropy sources exhibit intrinsic resilience to advanced modeling attacks [86]. PUFs based on spin-orbit torque MRAM (SOT-MRAM) or array-coupled structures

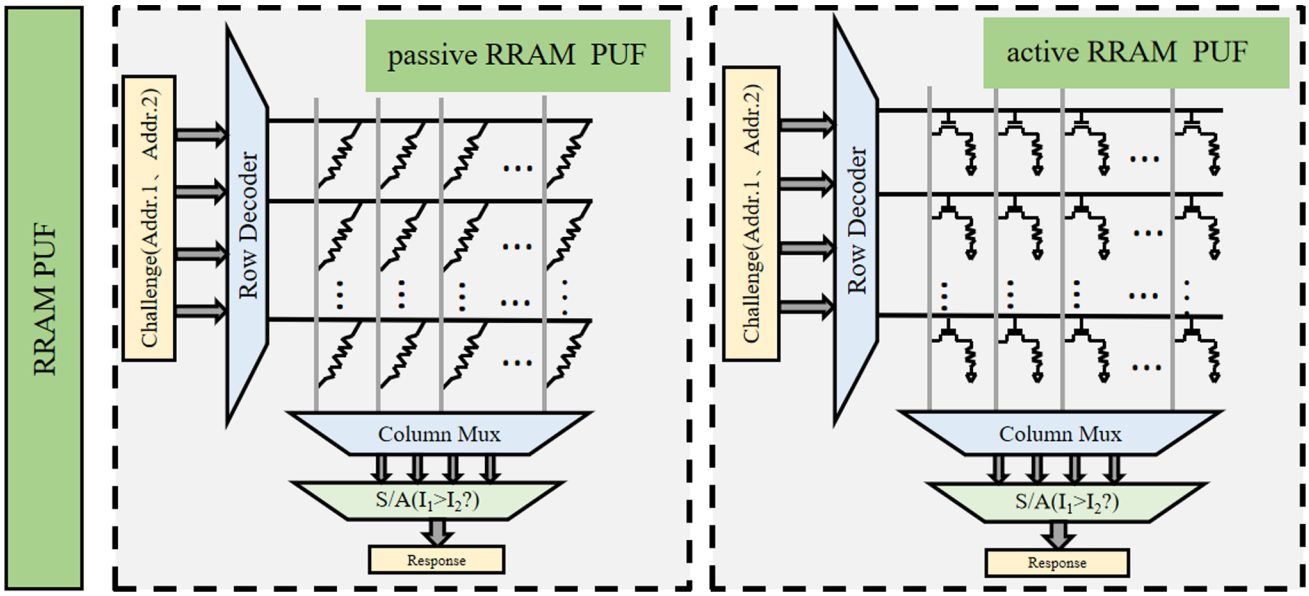


Fig. 6 The two fundamental structures of RRAM-based PUFs: passive RRAM devices and active RRAM

introduce nonlinear mapping and multi-dimensional coupling, reducing the prediction accuracy of machine learning models to near-random guessing and significantly weakening the effectiveness of deep learning attacks [91].

SOT-MRAM typically uses a 2T1R structure, and its high endurance allows frequent reconfiguration of the key space in the array. By combining pulses of different polarities, multiple variable CRPs can be generated over the operating temperature range, enabling the PUF to evolve from a traditional static fingerprint into a dynamically updatable security primitive, and thereby greatly enhancing its security. SOT-based PUFs offer excellent cycle reliability and support dynamic reconfiguration, but at the cost of more complex circuits (2T1R cell) and higher write current requirements [88].

3. Energy Efficiency and Area Issues

Compared with SRAM/DRAM PUFs, the nonvolatile nature of MRAM PUFs eliminates refresh power. By introducing sleep modes and optimizing write operations, the static power consumption can be reduced to the pW level [89]. Advanced spintronic devices such as SOT-MRAM feature separated read and write current paths, which can significantly improve write efficiency. Decoupling the read and write paths not only reduces write energy but also mitigates thermal disturbance and lower write-error probability, achieving approximately 70% lower reconfiguration power consumption [92].

Overall, MRAM PUFs outperform conventional CMOS PUFs in terms of low power consumption, high endurance, and integration density [93], but require precise sensing circuits and calibration schemes to ensure consistent performance. The choice of sensing circuit affects the power consumption, reliability, and implementation complexity of the PUF, requiring a trade-off between accuracy and overhead. Both current-mode and voltage-mode sense amplifiers can be applied to MRAM cells. For current-mode sensing, the MTJ read current is compared with a reference current and this produces a voltage output through transconductance amplification, offering good noise tolerance for low-voltage differences. For multi-level MTJ designs, current sensing provides more uniform sensing margins than voltage-mode sensing [94]. For voltage-mode sensing, the MTJ terminal voltage is compared with a reference voltage, featuring simpler circuitry but higher sensitivity to DC offsets. Wang et al. adopted a differential pair strategy: each data bit is stored by a complementary MTJ pair and read out via a differential amplifier, significantly improving noise immunity and effectively protecting against side-channel eavesdropping [88]. It should be noted that although MTJ array-based PUF architectures are simple and stable in output, they impose high requirements on sensing precision and are sensitive to temperature variations. In addition, peripheral circuits such as driver transistors and decoders are required to support write pulses. In the 1T1R structure, a single transistor is shared for read and write operations, whereas the 2T1R structure separates the read and write paths [88].

4.9 FRAM PUFs

Ferroelectric memories have emerged as one of the most promising candidates for general-purpose memory technologies due to their unique combination of nonvolatility, compatibility with CMOS processes, high endurance, and a high integration. The discovery of ferroelectricity in doped

HfO_2 was first reported by Boscke et al. [95], marking a significant milestone in this domain. Since then, various architectures, including 1T1C, FeFET, and FTJ [96], have been developed based on HfO_2-based ferroelectric memories. These structures have demonstrated substantial potential in diverse applications, including PUFs, further showcasing their versatility and importance in secure memory technologies.

4.9.1 Entropy sources

The spontaneous polarization of the ferroelectric thin film is derived from the electrical dipole moment created by the non-coincidence of positive and negative charge centers within the ferroelectric material's crystal cells, as illustrated in Fig. 7. The ferroelectric domains within the thin film are randomly distributed. Figure 7a, b illustrate the fully polarized states in positive and negative directions (Point A and Point B, respectively), while Fig 7c corresponds to the intermediate state during ferroelectric domain switching. Taking the positively polarized state as an example, gradually decreasing the strength of the positive electric field and even increasing the strength of the negative electric field leads to domain flipping when the coercive field is reached. This switching process is inherently probabilistic [97, 98]. The entropy source of ferroelectric devices for PUF applications fundamentally originates from process-induced variability and the intrinsic physical randomness of ferroelectric materials, which are collectively reflected in the inherent differences in the coercive electric field or threshold voltage of the devices.

4.9.2 Development trends

PUFs based on the three typical structures of ferroelectric devices have gradually evolved toward distinct yet, parallel development trends due to their differences in their structural characteristics and application requirements.

In **1T1C ferroelectric memory**, the transistor is equivalent to a switch and is generally fabricated using conventional CMOS processes. Therefore, research on 1T1C ferroelectric MemPUFs focuses on the ferroelectric capacitor. A ferroelectric capacitor is a metal/ferroelectric film/metal sandwich structure. Meihar et al. demonstrated a multiple-time programmable PUF [99] by extracting the P-V curves of fabricated capacitors and simulating them with SPICE. Although 1T1C ferroelectric memory shares structural similarities with DRAM, the intrinsic polarization switching mechanism of ferroelectric materials necessitates destructive read operations. Their non-volatile storage properties fundamentally differ from DRAM. These characteristics prevent direct translation of existing capacitor PUF which use DRAMs architectures to the 1T1C ferroelectric memory platforms.

Ferroelectric field-effect transistors (FeFETs) are transistors that employ ferroelectric materials as the gate dielectric, where the remanent polarization of the ferroelectric layer modulates channel conductance. Compared with the conventional 1T1C structure, FeFETs possess inherent three-dimensional integration capability, which is more favorable for high-density security chip implementation. During thin-film deposition, etching, and other process steps, inevitable defects and process variations lead to discrete distributions of device parameters such as threshold voltage and current, providing a stable and abundant physical entropy source for PUFs.

Taking advantage of the transistor structure, FeFET PUFs can directly replace CMOS devices in conventional PUFs, facilitating hardware security designs with low power consumption and a large CRP space. Breyer et al. first proposed and verified the implementation of AND and OR logic gates using a single FeFET, laying the foundation for subsequent reconfigurable security circuits [100, 101]. On this basis, Ramanujam et al. designed a multiplexer using the inherent parameter variations of FeFETs to construct an arbiter PUF [102]. Guo et al. further proposed two low-power reconfigurable PUF architectures: single transistor cell (1T/C) and dual transistor cell (2T/C) [103]. Specifically, the 1T/C structure features a compact cell layout with integrated counting and reference adjustment circuits, making it suitable for area-constrained scenarios with relatively sufficient computing resources. In contrast, the 2T/C architecture is more suitable for applications with relaxed area requirements and demand for fast reconfiguration. Simulations and experimental results show that both structures exhibit excellent uniqueness and reproducibility.

Despite outstanding performance, the practical application of FeFETs is limited by endurance degradation caused by repeated polarization switching. Studies have shown that the co-optimization of the ferroelectric (FE) layer and interfacial layer (IL) using a HAO/Al_2O_3 stacked structure can effectively suppress fatigue effects and improve device endurance [104].

Current research on high-performance FeFET PUFs generally focuses on the co-optimization of high entropy, high security, and high reliability. Shao et al. proposed a complementary symmetric FeFET PUF, which improves resistance to machine learning attacks by breaking the linear mapping relationship between challenges and responses [105]. Meanwhile, the small loop programming characteristic of ferroelectric materials is used to enhance the PUF entropy source, and a window comparator is adopted on the bit line to replace the conventional comparator, significantly improving the stability of bit output. Xu et al. implemented hardware-level information encryption by constructing a PUF-like array combined with an XOR encryption mechanism [106]. Even if stored data is leaked, the original information cannot

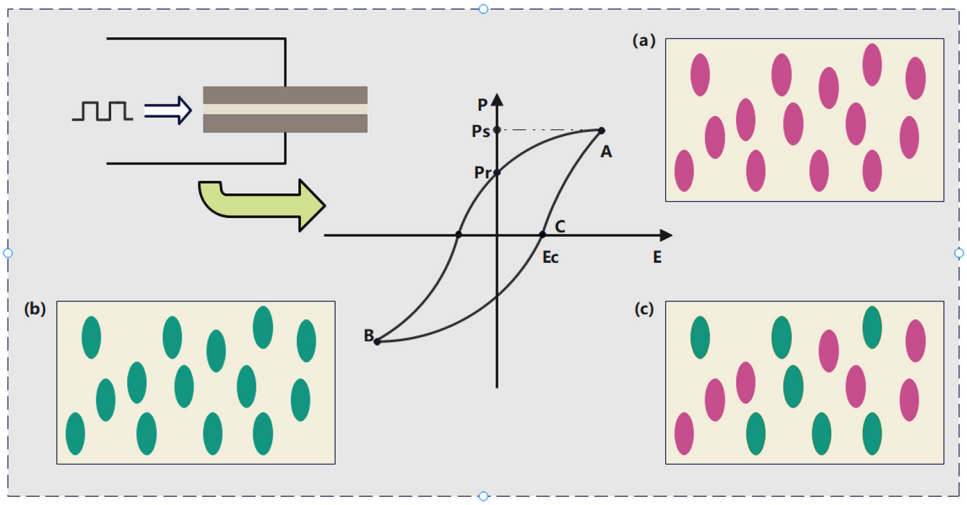


Fig. 7 **a** Fully polarized domains in the positive direction. **b** Fully polarized domains in the negative direction. **c** Ferroelectric domain flip with randomness

be recovered without a unique key. To address the issues of high power consumption and large area overhead in conventional PUFs, Lim et al. proposed a novel low-power FeFET PUF that reduces operating power consumption by leveraging the high on/off current ratio of the device [107]. A write-back mechanism is also employed to improve PUF stability without introducing additional area or power costs. Recently, Li et al. implemented a strong PUF with both high reconfigurability and ultra-low power consumption by exploiting the C2C variation of FeFETs and charge-domain computing, which was verified in 28 nm node process[108].

Ferroelectric tunnel junctions (FTJs) are resistive devices based on the tunneling electroresistance effect of ferroelectric materials, in which information storage is realized by regulating the polarization state of the ferroelectric layer via tunneling current. FTJs exhibit obvious similarities to FeFETs and passive RRAM in both structure and operating mechanism: all of them can be adapted to CBA architectures and implement PUF functions through current accumulation and comparison. Mahmoodi et al. extended RRAM PUFs from single devices to large scale crossbar arrays [83], and this design strategy provides an important reference for the array implementation of FTJ and FeFET-based PUFs. In addition, FTJs operate at the nanoampere (nA) current level, with power consumption much lower than that of conventional RRAM-based devices, making them highly promising in low-power hardware security applications.

FTJ PUFs utilize tunneling currents to regulate the polarization state of the ferroelectric layer [109]. Current research on FTJ PUFs mainly focuses on diverse application scenarios. Kim et al. proposed a PUF based on FTJ arrays [110]. Through simulations of arrays with varying sizes, it demonstrates that the system remains stable even at scaled-down dimensions. This opens up potential avenues for commercializing this new type of non-volatile MemPUFs. Lim et al. advanced this concept by connecting self-rectifying FTJs in series [111], with the two cells rectified in opposite directions. By configuring this array structure, they realized dual-mode operation for content-addressable memory (CAM) and PUF. Compared to separate implementations of CAM and PUF structures, this approach achieves the smaller area and lower search energy consumption while exhibiting the highest randomness. However, these improvements are accompanied by degraded reliability characteristics.

4.10 PRAM PUFs

4.10.1 Entropy sources

Similar to other emerging MemPUFs, the entropy source of PRAM PUFs is a composite system driven by both static process variations and dynamic programming variations.

Static process variations originate from random fluctuations in geometric dimensions introduced during manufacturing steps such as photolithography and thin-film deposition. These variations are fixed after chip fabrication and provide the fundamental physical fingerprint for the PUF [67]. The magnitude of variation can be tuned by adjusting the cell aspect ratio (L/W) and programming pulse parameters, including pulse width and edge steepness [112].

Dynamic programming variations arise from stochastic physical processes during phase-change memory operations, manifesting as random atomic structural rearrangements in the RESET and SET processes. This leads to resistance fluctuations across different programming cycles [112].

4.11 Development trends

While programming variations enhance the richness of the entropy source, they also introduce several engineering challenges, mainly including reliability issues and energy consumption overheads. These factors need to be carefully

balanced and optimized during PUF design and system integration.

1. Reliability Issues

By exploiting the virgin state of embedded PRAM employing CRP selection techniques, reliability superior to that of RRAM PUFs can be achieved [113]. However, the material undergoes spontaneous crystallization at elevated temperatures, since PRAM relies on switching between crystalline and amorphous phases to produce distinct resistance states. Such variations significantly degrade the reliability of the corresponding PUF designs; for instance, performance may degrade under $\pm 10\%$ power supply fluctuations and across the temperature range of $-40°C$ to $+70°C$, requiring the integration of error correction techniques [45] or compensation schemes [114].

2. Security Issues

PRAM PUFs further enhance robustness against modeling attacks by introducing nonlinear circuit structures, obfuscation of CRP behavior, and dynamic reconfiguration mechanisms based on physical randomness sources such as stochastic switching behavior. Meanwhile, keyless security architectures using the intrinsic process variations and random entropy sources of PRAM [67] lay the fundamental groundwork for constructing fully keyless hardware security primitives.

3. Energy Efficiency

Programming operations, especially RESET operation, generally incur high power consumption. Generating keys through read and write operations of existing memory cells avoids the area and energy overhead of dedicated security circuits. Additionally, hybrid memory architectures can reduce energy consumption by decreasing the number of write operations and optimizing scheduling strategies [115].

Benefiting from mature programming schemes, PRAM and DRAM hybrid memory architecture achieves an energy-delay product that is only 40% of the all-DRAM baseline under typical workloads.[116]. Adopting a frequent-value storage architecture also helps reduce the write energy consumption [117]. Furthermore, write latency and endurance can be improved via the Flip-N-Write microarchitectural technique, which replaces direct write operations with read-modify-write operations [118].

Despite the existing research, the scarcity of high-quality results in recent years indicates that PRAM PUFs have been phased out of mainstream research. Temperature sensitivity is an inherent and irreparable drawback [91]. The current trend for PUFs is toward low power consumption, high stability, high entropy, and aging resistance, whereas the energy consumption of programming and erasure in PRAM (in pJ/bit) is several times higher than that of ferroelectric memories (in fJ/bit) [119].

5 Indicators for MemPUFs

5.1 Measurement indicators for PUFs

PUFs generate random responses by leveraging the random physical attributes inherent in the chip, which arise due to variations in the manufacturing processes, even among devices produced under identical conditions. Therefore, the same PUF entity subjected to different challenges will produce distinct responses. In an ideal scenario, the same PUF entity will reproducibly yield identical responses to identical challenges. To ensure robust functionality, the response sequences must exhibit the following properties.

5.2 Uniqueness of PUFs

Uniqueness refers to the property that each chip produces a distinct response, effectively serving as a fingerprint that can uniquely identify the chip under ideal conditions. This characteristic is typically evaluated using the Hamming distance (HD). The HD measures the dissimilarity between two equal-length strings, specifically counting the number of positions at which the corresponding elements differ. It is mathematically defined by the equation in Eq. 1.

$$HD(R_p, R_q) = \sum_{s=1}^{n} R_p[s] \oplus R_q[s]. \tag{1}$$

Here, R_p and R_q represent two responses of length n. In the same operating environment, when different chips of the same PUF type are presented with the same challenge, their output responses should ideally differ. In an ideal scenario, the probability of two distinct chips producing identical responses is negligible. However, it is not impossible for identical responses to occur, particularly when the response length is insufficient. For k PUF devices, the Hamming distance between responses can be calculated as given in Eq. 2 [120], with the ideal value of the Hamming distance being 50%.

$$HD_{inter} = \frac{2}{k(k-1)} \sum_{i=1}^{k-1} \sum_{j=i+1}^{k} \frac{HD(R_i, R_j)}{n} \times 100\%. \tag{2}$$

5.3 Reliability of PUFs

Reliability is defined as the degree of similarity in the responses of the same device under identical challenges. In practical experiments, the responses of PUFs are susceptible to environmental factors. External influences such as temperature fluctuations and variations in read-out voltage can introduce errors into the output response of the same chip

under the same excitation signal. These errors should remain within a correctable range to ensure reliable operation.

Ideally, when environmental conditions remain constant, the intra-chip Hamming distance (HD_{intra}) should be 0%. To evaluate HD_{intra}, the same input challenge is applied m times, and the corresponding responses are extracted. R_{ref} denotes the reference response of length n. The calculation of HD_{intra} [120] is expressed as:

$$HD_{intra} = \frac{1}{m}\sum_{i=1}^{m}\frac{HD(R_{ref}, R_i)}{n} \times 100\%. \tag{3}$$

In a key generator, the parameters of the error correction code need to be selected according to the reliability of the original data in the PUF to ensure that the error correction code can correct errors caused by insufficient reliability.

5.4 Randomness of PUFs

The randomness of a PUF, often referred to as its uniformity, quantifies the balance between 0 and 1 in the response sequence generated by the PUF chip, with an ideal target of 50%. This metric can be evaluated using the response distribution test (Eq. 4), where $R[j]$ represents the value of the j bit in the output response. Achieving high randomness is critical for security, as it hinders an attacker's ability to computationally predict the PUF responses. Conversely, if the randomness deviates significantly towards 0% or 100%, the responses predominantly consist of "1" or"0", making the PUF susceptible to attacks.

$$Uniformity = \frac{1}{n}\sum_{j=1}^{n} R[j] \times 100\%. \tag{4}$$

The randomness of a PUF can also be analyzed through the lens of information theory. Shannon's formula provides a means to quantify entropy as a measure of information [121], expressed as follows:

$$H = -\sum_{i=0}^{n} P_i \log_2 P_i. \tag{5}$$

In a binary system, $H = -P_0 \log_2 P_0 - P_1 \log_2 P_1$, the source of signal entropy is maximized when 0 and 1 occur with equal probability. Joint Shannon entropy effectively evaluates PUF randomness. When minimum entropy is employed to quantify the entropy of a stochastic process under worst-case conditions (Eq. 6), the ideal value is 1. A minimum entropy of 1 represents the highest level of stochasticity.

$$H_{min} = -log_2 max(P_{(i=0)}, P_{(i=1)}). \tag{6}$$

In addition, NIST testing [122] can be utilized to evaluate the randomness of PUFs. PUFs that successfully pass NIST tests demonstrate robust security.

5.5 Performance analysis

To comprehensively clarify the practical deployment potential of emerging MemPUFs, we systematically evaluated the performance metrics of memory PUFs in recent years across core dimensions including reliability, security, and energy efficiency, and summarized the key comparative results in Table 2. Meanwhile, their potential advantages in IoT applications are further analyzed in Table 3.

The NIST Statistical Test Suite for Random Numbers is the standardized benchmark for evaluating the randomness of PUF-generated keys, and all candidate schemes must pass this test to verify their statistical properties. Meanwhile, ML modeling attacks have become the primary security threat to emerging MemPUFs. Thorough anti-attack evaluations must be conducted at the design stage to ensure sufficient security even under limited CRP leakage. Our main conclusions are as follows:

1. In terms of energy efficiency, device selection is the decisive factor. Emerging MemPUFs exhibit non-volatility, eliminating the need for periodic refresh, and feature lower static power consumption than conventional MemPUFs. In emerging memory technologies, hafnium-based ferroelectric devices (including FTJs [110], and FeFETs [105, 108]) possess intrinsic low power characteristics, and PUFs constructed from these devices significantly outperform RRAM and MRAM counterparts in both programming and readout energy consumption. Furthermore, complex peripheral circuits or ECCs introduced to improve sensing accuracy and reliability will significantly offset the inherent energy advantages of these devices. Optimizing the scheme during the PUF generation process (e.g., using charge domain entropy sources [108]) is more effective in reducing overall system energy consumption than relying solely on post-processing error correction techniques.

2. Reliability remains the primary bottleneck hindering the practical application of all emerging MemPUFs. Entropy sources based on novel physical mechanisms are highly sensitive to environmental conditions (temperature, supply voltage) and device aging, leading to response bit flips over time and under varying operating conditions. Although the aforementioned methods including cell screening, majority voting, error correction coding, and temperature compensation can improve reliability to a certain extent, these solutions inevitably increase hardware overhead. Therefore, achieving highly reliable PUF designs over wide temperature and voltage ranges without significantly sacrificing performance and area is the core scientific problem that urgently needs to be addressed in this field.

Table 2 Performance comparison of MemPUFs

	Entropy Source	Technology	Reliability	ML Attack	NIST	Uniqueness	Energy Efficiency	Area per cell	Temperature range(°C)
NC'21 [127]	RRAM	Hybrid organic electronic materials	~0% @85°C	54%	Passed	50.01%	51.2fJ/bit	~10000μm^2	–
EDL'21 [110]	FTJ	Switching current	3.8% @RT	–	Not tested	~49%	0.11fJ/bit	~0.0625μm^2	0–85
IEDM'23 [105]	FeFET	Minor loop current domain	1.7% @100°C	~50%	Passed	50.01%	3fJ/bit	~2500μm^2	25–100
Electronics'24 [128]	SRAM	Timing control scheme	4.86%@RT	–	Not tested	49.49%	7000fJ/bit	–	–
VLSI'24 [79]	RRAM	28nm + Auxiliary circuits/ Calibration	0.01%@25 °C	~50%	Passed	50.01%	23fJ/bit	~0.05265μm^2	−40–125
Sci. Rep'24 [93]	MRAM	Selection circuit	2.52% @-25°C	~49.87%	Passed	~49.96%	9.57fJ/bit	7.69μm^2	−25–100
NC'25 [108]	FeFET	28nm + Charge domain	0.7%@85 °C	52.1%	Passed	49.98%	1.89fJ/bit	1μm^2	25–85
TCAD'25 [129]	RRAM	90nm + 2 Entropy source	>10%@25 °C,40cycles	–	Passed	49.97%	–	~0.0064μm^2	−25-125
Engineering'25 [91]	SOT-MRAM	180nm + CIM	~0%@100°C	~50%	Passed	49.89%	–	0.315μm^2	−40–100

"−" means the scheme does not have this indicator, "RT" stands for room temperature

Table 3 Comparison of MemPUFs for IoT

Memory	Entropy Level	Reliability	Reconfigurability	Power Profile	IoT Implication
SRAM	Medium	High	No	Ultra-low (no write)	Lightweight key generation in MCUs
DRAM	Medium	Medium	Limited	High (refresh required)	Low-cost authentication in commodity devices
Flash	Medium	High	Limited	Medium	Secure ID storage in embedded systems
RRAM	High	Medium	Strong	Low	In-memory security and neuromorphic IoT
MRAM	High	High	Strong	Ultra-low standby	Neuromorphic computing schemes in edge devices
FRAM	High	High	Moderate	Ultra-low	Battery-free and energy-harvesting IoT
PRAM	High	Medium	Moderate	High (write-intensive)	Stable secure storage under constant temperature

6 Future research outlook

Emerging MemPUFs have evolved from traditional static hardware identification technologies into dynamically reconfigurable core security primitives, offering a highly promising solution to hardware security challenges in the IoT era. In addition to providing richer composite entropy sources, emerging memories have also spawned two transformative technologies: MLC and CIM, opening up entirely new directions for the future development of MemPUFs.

6.1 Multi-level storage and PUFs

MLC is one of the most fundamental distinguishing features of emerging non-volatile memories compared to conventional binary memories, and also the most promising yet underexplored technical direction in current MemPUF research. Virtually all conventional MemPUFs generate responses based on binary states, while emerging memories can achieve multi-level storage through precise modulation of their physical states [130, 131]. We argue that this continuously tunable analog characteristic brings a revolutionary new dimension to PUF design:

1. Multi-level storage can drastically improve entropy source density and the number of CRPs [132, 133]. An MLC capable of storing n bits of information can theoretically generate n bits of PUF response per cell, meaning that with the same chip area, the entropy source density of multi-level storage PUFs is n times that of binary PUFs.

2. Multi-level storage enhances resistance to machine learning attacks. The CRP mapping of binary PUFs is often linear or approximately linear, making them highly vulnerable to modeling attack by machine learning models such as neural networks. In contrast, the responses of multi-level storage PUFs are generated through quantization and combination of multiple continuous analog quantities, endowing their input–output relationships with inherent nonlinearity and complexity.

3. Multi-level storage provides a new degree of freedom for PUF reliability optimization. In binary PUFs, a bit error occurs as soon as the resistance of a cell crosses the reference resistance, leaving no room for error tolerance. Introducing an appropriate threshold window for decision-making can improve reliability [105]. In multi-level storage PUFs, however, the resistance distribution can be divided into multiple intervals by setting multiple reference levels, thereby significantly improving the fault tolerance of sensing operations.

Despite these significant advantages, multi-level storage may introduce numerous adverse effects on the performance, reliability, and circuit design of PUFs:

1. As the number of storage levels increases, the precision and complexity requirements for sensing circuits grow exponentially. Current-mode sensing offers better noise tolerance but higher power consumption, while voltage-mode sensing has simpler circuitry but is more sensitive to offset voltage. This trade-off may become even more extreme in multi-level storage PUFs. Furthermore, multi-level storage requires precise control over the physical states of devices, which imposes extremely high demands on programming circuitry.

2. Multi-level storage is far more sensitive to environmental disturbances and device aging than binary storage, which exacerbates the already existing reliability issues. In multi-level storage, minor variations in temperature, voltage, or device parameters are sufficient to cause state misclassification.

3. Although multi-level storage enhances PUF resistance to machine learning modeling attacks, it also introduces new attack surfaces. By injecting minute voltage or temperature perturbations, adversaries can easily alter the states of MLCs and thus disrupt the normal operation of the PUF.

6.2 Computing-in-memory and PUFs

We particularly highlight that the convergence of emerging memory-based Computing-in-Memory (CIM) and PUFs is the most promising research direction. CIM performs computation operations directly within the memory array, while PUFs inherently leverage the intrinsic physical properties of memories to implement security functions. Their inherent homology enables a dual-mode "computing-security" design: the same memory array can not only execute CIM tasks such as neural network inference and matrix operations, but also generate cryptographic keys, implement device authentication and perform data encryption using its inherent physical randomness.

At present, preliminary studies have explored the dual-mode design [91, 134], but this field remains in its infancy. Future research needs to deeply investigate the co-design methodologies of CIM and PUFs, resolve their conflicts in operation timing, voltage ranges and reliability requirements, and explore novel device structures and array architectures that can simultaneously optimize both computation performance and security performance.

7 Conclusion

Faced with the urgent demand for lightweight hardware security primitives in the IoT and edge computing, this paper systematically reviews the research context and technological evolution of emerging MemPUFs. From the dimensions of device types, entropy source mechanisms, performance metrics, and development trends, this paper establishes a comprehensive MemPUF technology lineage covering traditional memories and emerging memories such as RRAM,

MRAM, FRAM, and PRAM, and constructs a unified quantitative evaluation framework based on technology node normalization.

Through statistical analysis and horizontal comparison of representative works in the past five years, this paper reveals the decisive impact of device selection on the overall performance of PUFs, pointing out that hafnium-based ferroelectric devices exhibit significantly better energy efficiency than other emerging memory technologies. On this basis, this paper deeply analyzes the dual effects of MLC technology on the entropy density, attack resistance, and reliability of MemPUFs, demonstrating that while MLC improves entropy source density and input–output nonlinearity, it also significantly increases the complexity of sensing circuits and environmental sensitivity. Furthermore, this paper prospectively discusses the "computing-security" integration architecture that deeply merges CIM and PUFs, clarifying the co-design challenges and feasible technical solutions in terms of operation timing, voltage ranges, and reliability constraints.

This work fills the gap in the systematic review of entropy sources and development trends in the field of emerging nonvolatile memory PUFs. The proposed development paradigm can provide theoretical references and technical guidelines for the design of next-generation low-power and high-security hardware security primitives for edge devices. In the future, relying on the collaborative optimization of MLC and CIM, it is expected to realize integrated secure chips with high entropy, high reliability, high energy efficiency, and resistance to machine learning attacks.

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References

1. Sinha, S., State of IoT 2025: Number of connected IoT devices growing 14% to 21.1 billion globally. (2025). https://iot-analytics.com/number-connected-iot-devices/
2. Herder, C., Yu, M.D., Koushanfar, F., Devadas, S.: Physical unclonable functions and applications: a tutorial. Proc. IEEE **102**(8), 1126–1141 (2014). https://doi.org/10.1109/JPROC.2014.2320516
3. Pappu, R., Recht, B., Taylor, J., Gershenfeld, N.: Physical one-way functions. Science (2002). https://doi.org/10.1126/science.1074376
4. Gassend, B., Clarke, D., Van Dijk, M., Devadas, S.: Delay-based circuit authentication and applications. ACM Symp. Appl. Comput, Proc. (2003). https://doi.org/10.1145/952532.952593
5. Zerrouki, F., Ouchani, S., Bouarfa, H.: A survey on silicon PUFs. J. Syst. Archit. **127**, 102514 (2022). https://doi.org/10.1016/j.sysarc.2022.102514
6. Dai, S., Wei, J., Zhang, F.: Memory leakage-resilient secret sharing schemes. Sci. China Inf. Sci. **58**(11), 1–9 (2015). https://doi.org/10.1007/s11432-015-5385-8
7. Bing, L., Chen, S., Kai, W.: An improved PAAs countermeasure based on permutation tables and latch PUF. Sci. China Inf. Sci. **63**(139104), 24 (2020). https://doi.org/10.1007/s11432-018-9540-y
8. Park, S., Na, H., Choi, J., Ismail, M., Mahata, C., Ryu, D., Kim, S., Lee, J.-K., Yu, J., Kim, S.: Dual-state conversion for high-entropy and reconfigurable resistive memory-based physically unclonable functions. J. Mater. Sci. Technol. **266**, 38–47 (2026). https://doi.org/10.1016/j.jmst.2025.11.025
9. Mohammadinodoushan, M., Cambou, B., Philabaum, C., Hely, D., Booher, D. D.: Implementation of password management system using ternary addressable PUF generator, 2019 16th Annual IEEE International Conference on Sensing, Communication, and Networking (SECON), Boston, MA, USA, (2019) pp. 1–8. https://doi.org/10.1109/SAHCN.2019.8824792
10. Mahalat, M.H., Karmakar, D., Mondal, A., Sen, B.: PUF based secure and lightweight authentication and key-sharing scheme for wireless sensor network. J. Emerg. Technol. Comput. Syst. **18**(9), 78 (2022). https://doi.org/10.1145/3466682
11. Barbareschi, M., Casola, V., Emmanuele, A., Lombardi, D.: A lightweight PUF-based protocol for dynamic and secure group key management in IoT. IEEE Internet Things J. **11**(20), 32969–32984 (2024). https://doi.org/10.1109/JIOT.2024.3418207
12. Chatterjee, U., Govindan, V., Sadhukhan, R., Mukhopadhyay, D., Chakraborty, R.S., Mahata, D., Prabhu, M.M.: Building PUF based authentication and key exchange protocol for IoT without explicit CRPs in verifier database. IEEE Trans. Depend. Secure Comput. **16**(3), 424–437 (2019). https://doi.org/10.1109/TDSC.2018.2832201
13. Zheng, Y., Liu, W., Gu, C., Chang, C.-H.: PUF-based mutual authentication and key exchange protocol for peer-to-peer IoT applications. IEEE Trans. Dependable Secure Comput. **20**(4), 3299–3316 (2023). https://doi.org/10.1109/TDSC.2022.3193570
14. Nakkar, M., AlTawy, R., Youssef, A.: Lightweight group authentication scheme leveraging Shamir's secret sharing and PUFs. IEEE Trans. Network Sci. Eng. **11**(4), 3412–3429 (2024). https://doi.org/10.1109/TNSE.2024.3373386

15. Gao, Y., Al-Sarawi, S.F., Abbott, D.: Physical unclonable functions. Nat. Electron. **3**(2), 81–91 (2020). https://doi.org/10.1038/s41928-020-0372-5
16. Mahalat, M. H., Anwar, M. S., Kumar, S., Sen, B.: Physically unclonable function (PUF) a lightweight security primitive for IoT: Scope and challenges, In: Proceeding International Conference Machine Learning Computer System Security (MLCSS), pp. 368-373 (2022). https://doi.org/10.1109/MLCSS57186.2022.00073
17. Shamsoshoara, A., Korenda, A., Afghah, F., Zeadally, S.: A survey on physical unclonable function (PUF)-based security solutions for Internet of Things. Comput. Netw. **183**, 107593 (2020). https://doi.org/10.1016/j.comnet.2020.107593
18. Lata, K., Cenkeramaddi, L.R.: FPGA-based PUF designs: a comprehensive review and comparative analysis. Cryptography (2023). https://doi.org/10.3390/cryptography7040055
19. Zhang, J., Qu, G., Lv, Y., Zhou, Q.: A survey on silicon PUFs and recent advances in ring oscillator PUFs. J. Comput. Sci. Technol. **29**(4), 664–678 (2014). https://doi.org/10.1007/s11390-014-1458-1
20. Hemavathy, S., Bhaaskaran, V.S.K.: Arbiter PUF-A review of design, composition, and security aspects. IEEE Access **11**, 33979–34004 (2023). https://doi.org/10.1109/ACCESS.2023.3264016
21. Guajardo, J., Kumar, S. S., Schrijen, G.-J., Tuyls, P.: FPGA intrinsic PUFs and their use for IP protection, In: Proceeding 9th International Workshop Cryptography Hardware Embedded System
22. Maes, R., van der Leest, V., van der Sluis, E., Schrijen, G.-J.: Secure key generation from biased PUFs, In: Proceeding 17th International Workshop Cryptography Hardware Embedded System(CHES), Saint-Malo, France, (2015) pp. 517–534
23. Sutar, S., Raha, A., Raghunathan, V.: Memory-based combination PUFs for device authentication in embedded systems. IEEE Trans. Multi-Scale Comput. Syst. **4**(4), 793–810 (2018). https://doi.org/10.1109/TMSCS.2018.2885758
24. Bulens, P., Standaert, F.-X., Quisquater, J.-J.: How to strongly link data and its medium: the paper case. IET Inf. Secur. **4**(3), 125–136 (2010). https://doi.org/10.1049/iet-ifs.2009.0032
25. Hammouri, G., Dana, A., Sunar, B.: CDs have fingerprints too, In: Proceeding International Workshop Cryptography Hardware Embedded System (CHES), Lausanne, Switzerland, pp. 348–362 (2009)
26. Tuyls, P., Schrijen, G.-J., škorić, B., van Geloven, J., Verhaegh, N., Wolters, R.: Read-proof hardware from protective coatings, In: Proceeding 8th International Workshop Cryptography Hardware Embedded System (CHES), Yokohama, Japan, pp. 369–383 (2006)
27. B. škorić, S. Maubach, T. Kevenaar, P. Tuyls,: Information-theoretic analysis of capacitive physical unclonable functions. J. Appl. Phys. **100**(2), 024902 (2006). https://doi.org/10.1063/1.2209532
28. Guajardo, J., B. škorić, P. Tuyls, S. S. Kumar, T. Bel, A. H. M. Blom, G. J. Schrijen,: Anti-counterfeiting, key distribution, and key storage in an ambient world via physical unclonable functions. Inf. Syst. Front. **11**(1), 19–41 (2009). https://doi.org/10.1007/s10796-008-9142-z
29. Ren, P., Xue, Y., Jing, L., Zhang, L., Wang, R.: A strong physical unclonable function with machine learning immunity for Internet of Things application. Sci. China Inf. Sci. **67**(1), 112404 (2024). https://doi.org/10.1007/s11432-022-3722-8
30. Yao, L., Liang, H., Huang, Z., Jiang, C.: A lightweight configurable XOR RO-PUF design based on Xilinx FPGA, In: Proceeding 4th IEEE International Conference Electronic Technology, Chengdu, China, pp. 83-88 (2021). https://doi.org/10.1109/ICET51757.2021.9451016
31. Lim, D., Lee, J.W., Gassend, B., Suh, G.E., van Dijk, M., Devadas, S.: Extracting secret keys from integrated circuits. IEEE Trans. Very Large Scale Integr. (VLSI) **13**(10), 1200–1205 (2005). https://doi.org/10.1109/TVLSI.2005.859470
32. Gassend, B., Clarke, D., van Dijk, M., Devadas, S.: Silicon physical random functions, in Proceeding 9th ACM Conference Computer Communication Security, Washington, DC, USA, pp. 148-160 (2002). https://doi.org/10.1145/586110.586132
33. Gu, C., Hanley, N., O'Neill, M.: Improved reliability of FPGA-based PUF identification generator design. ACM Trans. Reconfig. Technol. Syst. **10**(3), 1–23 (2017). https://doi.org/10.1145/3053681
34. Suh, G. E., Devadas, S.: Physical unclonable functions for device authentication and secret key generation, In: *Proceeding 44th IEEE Design Automation Conference*, San Diego, CA, USA, pp. 9-14 (2007)
35. Fard, S.S., Kaveh, M., Mosavi, M.R., Ko, S.B.: An efficient modeling attack for breaking the security of XOR-Arbiter PUFs by using the fully connected and long-short term memory. Microprocess. Microsyst. **94**, 104667 (2022). https://doi.org/10.1016/j.micpro.2022.104667
36. Gu, C., Cui, Y., Hanley, N., O'Neill, M.: Novel lightweight FF-APUF design for FPGA, In: Proceeding 29th IEEE design System-on-Chip Conference, Seattle, WA, USA, pp. 75-80 (2016). https://doi.org/10.1109/SOCC.2016.7905439
37. Xin, X., Kaps, J.-P., Gaj, K.: A configurable ring-oscillator-based PUF for Xilinx FPGAs, In: Proceeding 14th Euromicro Conference Digital System Design, Oulu, Finland, pp. 651-657 (2011). https://doi.org/10.1109/DSD.2011.88
38. Capovilla, J., Cortes, M., Araujo, G.: Improving the statistical variability of delay-based physical unclonable functions, In: Proceeding 28th Symposium Integrated Circuits System Design, Salvador, Brazil, pp. 1-7 (2015). https://doi.org/10.1145/2800986.2801010
39. Cui, Y., Wang, C., Liu, W., Yu, Y.: 'Low-cost configurable ring oscillator PUF with improved uniqueness, In: Proceeding 2016 IEEE International Symposium Circuits System, Montreal, QC, Canada, pp. 558-561 (2016) https://doi.org/10.1109/ISCAS.2016.7527301
40. Suzuki, D., Shimizu, K.: The glitch PUF: a new delay-PUF architecture exploiting glitch shapes, In: Proceeding 12th International Workshop Cryptography Hardware Embedded System (CHES), Santa Barbara, CA, USA, pp. 366-382 (2010). https://doi.org/10.1007/978-3-642-15031-9_25
41. Shimizu, K., Suzuki, D., Kasuya, T.: Glitch PUF,: Extracting information from usually unwanted glitches. IEICE Trans. Fundam. Electron. Commun. Comput. Sci. (2012). https://doi.org/10.1587/transfun.E95.A.223
42. Streit, F.J., Krüger, P., Becher, A., Wildermann, S.: Design and evaluation of a tunable PUF architecture for FPGAs. ACM Trans. Reconfig. Technol. Syst. **15**(1), 1–27 (2022). https://doi.org/10.1145/3491237
43. Li, B., Chen, S.: A dynamic PUF anti-aging authentication system based on restrict race code. Sci. China Inf. Sci. **59**(1), 1–12 (2016). https://doi.org/10.1007/s11432-015-5287-9
44. Holcomb, D.E., Burleson, W.P., Fu, K.: Power-up SRAM state as an identifying fingerprint and source of true random numbers. IEEE Trans. Comput. **58**(9), 1198–1210 (2009). https://doi.org/10.1109/TC.2008.212
45. Vijayakumar, A., Patil, V.C., Kundu, S.: On improving reliability of SRAM-based physically unclonable functions. J. Low Power Electron. Appl. **7**(1), 2 (2017). https://doi.org/10.3390/jlpea7010002
46. Li, J., Yang, T., Seok, M.: A technique to transform 6T-SRAM arrays into robust analog PUF with minimal overhead, In: Proceeding IEEE International Symposium Circuits System

(ISCAS), Baltimore, MD, USA, pp. 1-4 (2017). https://doi.org/10.1109/ISCAS.2017.8050630
47. Liu, K., Min, Y., Yang, X., Wang, H.: A 373-F2 0.21%-Native-BER EE SRAM Physically Unclonable Function With 2-D Power-Gated Bit Cells and V_SS Bias-Based Dark-Bit Detection, IEEE J. Solid-State Circuits, vol. 55, no. 6, pp. 1719–1732 (2020)
48. Fritzsch, C., Schulz, W., Rosenkranz, T., Ahr, P., Saavedra, A., Renkes, L., Lipps, J. C.: Hoffmann:SRAM-based PUF implementation and bias patterns on AMD Xilinx UltraScale+ FPGAs.In Proceeding 14th International Workshop Hardware Architectural Support Security Privacy (HASP),pp. 82–90, 2025, https://doi.org/10.1145/3768725.3768731
49. Simpson, E., Schaumont, P.: Offline hardware/software authentication for reconfigurable platforms, In: Proceeding 8th International Workshop Cryptography Hardware Embedded System (CHES), Yokohama, Japan, pp. 311-323 (2006). https://doi.org/10.1007/11894063_25
50. Kumar, S. S., Guajardo, J., Maes, R., Schrijen, G.-J., Tuyls, P.: The butterfly PUF protecting IP on every FPGA, In: Proceeding IEEE International Workshop Hardware-Oriented Security. Trust (HOST), Anaheim, CA, USA, pp. 67-70 (2008)
51. Tehranipoor, F., Karimian, N., Yan, W., Chandy, J. A.: DRAM-based intrinsic physically unclonable functions for system-level security and authentication, IEEE Trans. Very Large Scale Integrated (VLSI), vol. 25, no. 3, pp. 1085–1097 (2017)
52. Liu, W., Zhang, Z., Li, M., Liu, Z.: A trustworthy key generation prototype based on DDR3 PUF for wireless sensor networks. Sensors (Basel) **14**(7), 11542–11556 (2014). https://doi.org/10.3390/s140711542
53. Keller, C., Gürkaynak, F., Kaeslin, H., Felber, N.: In: Proceeding IEEE International Symposium Circuits System (ISCAS), Melbourne, VIC, Australia, pp. 2740-2743 (2014). https://doi.org/10.1109/ISCAS.2014.6865740
54. Hashemian, M. S., Singh, B., Wolff, F., Weyer, D.: A robust authentication methodology using physically unclonable functions in DRAM arrays, In: Proceding Design Automation Test Europe Conference Exhibition (DATE), Grenoble, France, pp. 647-652 (2015). https://doi.org/10.7873/DATE.2015.0308
55. DRAM-based PUF utilizing the variation of adjacent cells: E. Abulibdeh, L. Younes, B. Mohammad, K. Humood. IEEE Trans. Inf. Forensics Security **19**, 2909–2918 (2024). https://doi.org/10.1109/TIFS.2024.3354115
56. Lofstrom, K., Daasch, W. R., Taylor, D.: IC identification circuit using device mismatch, in Proceeding IEEE International Solid-State Circuits Conference (ISSCC), San Francisco, CA, USA, pp. 372-373 (2000)
57. Lee, J., Kim, M., Jeong, M., Shin, G.: A 20F2/bit current-integration-based differential NAND-structured PUF for stable and V/T variation-tolerant low-cost IoT security. IEEE J. Solid-State Circuits **57**(10), 2957–2968 (2022). https://doi.org/10.1109/JSSC.2022.3192903
58. Lee, J., Kim, M., Shin, G., Lee, K.: A 20F2 area-efficient differential NAND-structured physically unclonable function for low-cost IoT security, In: Proceeding 45th Europe Solid-State Circuits Conference (ESSCIRC), Cracow, Poland, pp. 1-4. (2019). https://doi.org/10.1109/ESSCIRC.2019.8902680
59. Bai, Y., Wu, L., Wu, X., Li, X.: PUF-based encryption method for IC cards on-chip memories. Electron. Lett. **52**(20), 1671–1673 (2016). https://doi.org/10.1049/el.2016.0984
60. Jia, S., Xia, L., Wang, Z., Lin, J.: Extracting robust keys from NAND flash physical unclonable functions, In: Proceeding 18th International Conference Information Security (ISC), Trondheim, Norway, pp. 437-454 (2015). https://doi.org/10.1007/978-3-319-23318-5_24
61. Sakib, S., Milenkovic, A., Rahman, M.T., Ray, B.: An aging-resistant NAND flash memory physical unclonable function. IEEE Trans. Electron Dev. **67**(3), 937–943 (2020). https://doi.org/10.1109/TED.2020.2968272
62. Guo, Z., Xu, X., Tehranipoor, M. M., Forte, D.: FFD: a framework for fake flash detection, In: Proceeding 54th ACM/EDAC/IEEE Design Automation Conference (DAC), Austin, TX, USA, pp. 1-6 (2017). https://doi.org/10.1145/3061639.3062249
63. Sakib, S., Milenković, A., Ray, B.: Flash-DNA: identifying NAND flash memory origins using intrinsic array properties. IEEE Trans. Electron Dev. **68**(8), 3794–3800 (2021). https://doi.org/10.1109/TED.2021.3087454
64. Sahay, S., Klachko, M., Strukov, D.: Hardware security primitive exploiting intrinsic variability in analog behavior of 3-D NAND flash memory array. IEEE Trans. Electron Dev. **66**(5), 2158–2164 (2019). https://doi.org/10.1109/TED.2019.2903786
65. Song, M. S., Ahn, S., Hwang, H., Kim, H.: Memcapacitor crossbar array with charge trapping layer for physical unclonable function in NAND flash architecture. In: Proceeding IEEE International Electron Devices Meeting (IEDM), San Francisco, CA, USA, pp. 1-4 (2023). https://doi.org/10.1109/IEDM45741.2023.10413842
66. Chuang, K. H., Bury, E., Degraeve, R., Kaczer, B.: A physically unclonable function using soft oxide breakdown featuring 0% native BER and 51.8 fJ/bit in 40-nm CMOS. IEEE J. Solid-State Circuits, vol. 54(10) 2765–2776 (2019). https://doi.org/10.1109/JSSC.2019.2920714
67. Zhang, L., Kong, Z.H., Chip-Hong Chang, A., Cabrini, C.: Exploiting process variations and programming sensitivity of phase change memory for reconfigurable physical unclonable functions. IEEE Trans. Inf. Forensics Security **9**(6), 921–932 (2014). https://doi.org/10.1109/TIFS.2014.2315743
68. Goyal, R., Porti, M., Crespo-Yepes, A., Nafria, M.: Physical unclonable functions based on the post-breakdown current of FDSOI nanowire transistors. IEEE Electron Device Lett. **46**(8), 1273–1276 (2025). https://doi.org/10.1109/LED.2025.3577746
69. Tisha, Z.T., Muldavin, J., Guin, U.: Exploring security solutions and vulnerabilities for embedded non-volatile memories. IEEE Symp. VLSI (ISVLSI) (2024). https://doi.org/10.1109/ISVLSI61997.2024.00072
70. Liu, R., Wu, H., Pang, Y., Qian, H., Yu, S.: Experimental characterization of physical unclonable function based on 1 kb resistive random access memory arrays. IEEE Electron Device Lett. **36**(12), 1380–1383 (2015). https://doi.org/10.1109/LED.2015.2496257
71. Wei, Z., Kanzawa, Y., Arita, K., Katoh, Y.: Highly reliable TaOx ReRAM and direct evidence of redox reaction mechanism. In: Proceeding IEEE International Electron Devices Meeting (IEDM), San Francisco, CA, USA, pp. 1-4 (2008). https://doi.org/10.1109/IEDM.2008.4796676
72. Chen, A.: Utilizing the variability of resistive random access memory to implement reconfigurable physical unclonable functions. IEEE Electron Device Lett. **36**(2), 138–140 (2015). https://doi.org/10.1109/LED.2014.238587
73. Pang, Y., Wu, H., Gao, B., Liu, R.: Design and optimization of strong physical unclonable function (PUF) based on RRAM array. In: Proceeding International Symposium VLSI Technology System Application (VLSI-TSA), Hsinchu, Taiwan, pp. 1-2 (2017). https://doi.org/10.1109/VLSI-TSA.2017.7942473
74. Pang, Y., Wu, H., Gao, B., Wu, D.: A novel PUF against machine learning attack: implementation on a 16 Mb RRAM chip. In: Proceeding IEEE International Electron Devices Meeting (IEDM), San Francisco, CA, USA, pp. 12.2.1-12.2.4 (2017). https://doi.org/10.1109/IEDM.2017.8268376
75. Nili, H., Adam, G.C., Hoskins, B., Prezioso, M.: Hardware-intrinsic security primitives enabled by analogue state and nonlinear conductance variations in integrated memristors. Nat. Electron. **1**(3), 197–202 (2018). https://doi.org/10.1038/s41928-018-0039-7

76. Mahmoodi, M. R., Nili, H., Strukov, D. B.: RX-PUF: low power, dense, reliable, and resilient physically unclonable functions based on analog passive RRAM crossbar arrays. In: Proceeding IEEE Symposium VLSI Technology, Honolulu, HI, USA, pp. 99-100 (2018). https://doi.org/10.1109/VLSIT.2018.8510624
77. Yang, J., Chen, D., Ding, Q., Fang, J.: A novel PUF using stochastic short-term memory time of oxide-based RRAM for embedded applications. In: Proceeding IEEE International Electron Devices Meeting (IEDM), San Francisco, CA, USA, (2020). https://doi.org/10.1109/IEDM13553.2020.9372050
78. Chen, A., Comprehensive assessment of RRAM-based PUF for hardware security applications. In: Proceeding IEEE International Electron Devices Meeting (IEDM), Washington, DC, USA, pp. 10.7.1-10.7.4. (2015).https://doi.org/10.1109/IEDM.2015.7409672
79. Cao, Y., Yang, H., Yang, J., Zhao, W.: A 67 F^2 reconfigurable PUF using 1T2R RRAM switching competition in 28nm CMOS with 5×10^{-9} bit error rate. In: Proceeding IEEE Symposium VLSI Technology Circuits, Honolulu, HI, USA, pp. 1-2 (2024)
80. Schultz, T., Jha, R., Casto, M., Dupaix, B.: Vulnerabilities and reliability of ReRAM based PUFs and memory logic. IEEE Trans. Reliab. **69**(2), 690–698 (2020)
81. Kim, J., Ahmed, T., Nili, H., Yang, J., Jeong, D.S., Beckett, P., Sriram, S., Ranasinghe, D.C., Kavehei, O.: A physical unclonable function with redox-based nanoionic resistive memory. IEEE Trans. Inf. Forensics Security **13**(2), 437–448 (2018). https://doi.org/10.1109/TIFS.2017.2756562
82. Yang, J., Lei, D., Chen, D., Li, J.: A machine-learning-resistant 3D PUF with 8-layer stacking vertical RRAM and 0.014% bit error rate using in-cell stabilization scheme for IoT security applications. In: Proceeding IEEE Int. Electron Devices Meeting (IEDM), San Francisco, CA, USA, (2020). https://doi.org/10.1109/IEDM13553.2020.9372107
83. Mahmoodi, M. R., Nili, H., Fahimi, Z., Larimian, S., Kim, H., Strukov, D.: Ultra-Low Power Physical Unclonable Function with Nonlinear Fixed-Resistance Crossbar Circuits. In: Proceeding International Electron Devices Meeting (IEDM), San Francisco, CA, USA, pp. 30.1.1–30.1.4. (2019). IEEE
84. Dieny, B., Speriosu, V.S., Parkin, S.S.P., Gurney, B.A., Wilhoit, D.R., Mauri, D.: Giant magnetoresistive in soft ferromagnetic multilayers. Phys. Rev. B: Condens. Matter **43**(1), 1297–1300 (1991). https://doi.org/10.1103/PhysRevB.43.1297
85. Moodera, J.S., Kinder, L.R., Wong, T.M., Meservey, R.: Large magnetoresistance at room temperature in ferromagnetic thin film tunnel junctions. Phys. Rev. Lett. **74**(16), 3273–3276 (1995). https://doi.org/10.1103/PhysRevLett.74.3273
86. Kang, J., Han, D., Lee, K., Ko, S., Koh, D., Park, C., Ahn, J., Yu, M.i, Pakala, M., Noh, S., Lee, H., Kwon, J., Kim, K. J., Park, J., Lee, S., Lee, J., Park, B. G.: Highly reliable magnetic memory-based physical unclonable functions. ACS Nano, vol. 18, no. 20. (2024). https://doi.org/10.1021/acsnano.4c00078
87. Criss, T., Valli, A. S. E., Li, N., Haas, A., Kent, A. D.: Magnetic tunnel junction as a real-time entropy source: Field-Programmable Gate Array based random bit generation without post-processing. J. Appl. Phys., vol. 139, no. 1, pp. 013903, (2026). https://doi.org/10.1063/5.0308636
88. Wang, M., Jiang, C., Wang, Z., Hou, Z., Zhang, Z., Zhao, Y., Liu, H., Zhao, W.: Temperature?resilient reconfigurable physical unclonable function driven by pulse modulation using CMOS?integrated spintronic chips, Adv. Sci., (2026) https://doi.org/10.1002/advs.74926
89. Chen, Z., Zhang, Y., Shen, C.: A reconfigurable MRAM PUF with high reliability. Microelectron. J., vol. 163, pp. 106745, (2025). https://doi.org/10.1016/j.mejo.2025.106745
90. Meng, H., Wu, Y., Zhou, S., Zou, T., Min, T., Wang, S.: A Highly Reliable and Anti-Cloning STT-MRAM Reconfigurable Physical Unclonable Function. In: Proceeding 2025 IEEE European Solid-State Electronics Research Conference (ESSERC), 08-11 (2025). https://doi.org/10.1109/ESSERC66193.2025.11214072
91. Zhang, X., Jiang, C., Yin, J., Zhu, D., Wang, S., Li, S., Zhang, Z., Du, A., Cai, W., Liu, H., Shi, K., Cao, K., Wang, Z., Zhao, W.: Experimental realization of physical unclonable function chip utilizing spintronic memories. Engineering **49**, 141–148 (2025). https://doi.org/10.1016/j.eng.2024.12.028
92. Zhang, S., Guo, X., Wang, Y., Wang, C., Wang, B., Wang, Z., "Robust and Efficient NAND-Like TST-MRAM with Parallel Write, Read Operations and Reconfigurable PUF Mode,",: 26th International Symposium on Quality Electronic Design (ISQED). San Francisco, CA, USA **2025**, 1–5 (2025)
93. Adel, M. J., Rezayati, M. H., Moaiyeri, M. H., Amirany, A., Jafari, K.: A robust deep learning attack immune MRAM-based physical unclonable function,Sci. Rep.,vol. 14, Art. no. 20649, (2024). https://doi.org/10.1038/s41598-024-71730-7
94. Vatankhahghadim, A., A. Sheikholeslami.: A Multi-level Cell for STT-MRAM with Biaxial Magnetic Tunnel Junction. In: Proceeding: IEEE International Symposium on Multiple-Valued Logic (ISMVL). Waterloo, ON, Canada **2015**, 158–163 (2015). https://doi.org/10.1109/ISMVL.2015.38
95. Böscke, T.S., Müller, J., Bräuhaus, D., Schröder, U., Böttger, U.: Ferroelectricity in hafnium oxide thin films. Appl. Phys. Lett. **99**(10), 102903 (2011). https://doi.org/10.1063/1.3634052
96. Majumdar, S.: Can ferroelectric tunnel junction be a game changer as eNVM and in neuromorphic hardware? APL Mach. Learn. **3**(2), 020902 (2025). https://doi.org/10.1063/5.0252822
97. Jiang, B., Zurcher, Jones, Gillespie, Lee.: Computationally efficient ferroelectric capacitor model for circuit simulation. In: Proceeding Symposium VLSI Technology, Kyoto, Japan, pp. 141-142 (1997).https://doi.org/10.1109/VLSIT.1997.623738
98. Mulaosmanovic, H., Ocker, J., Müller, S., Schroeder, U., Müller, J., Polakowski, P., Flachowsky, S., Bentum, R.V., Mikolajick, T., Slesazeck, S.: Switching kinetics in nanoscale hafnium oxide based ferroelectric field-effect transistors. ACS Appl. Mater. Interfaces. **9**(4), 3792–3798 (2017). https://doi.org/10.1021/acsami.6b13866
99. Meihar, P., Rowtu, S., Lashkare, S., Ganguly, U.: A ferroelectric differential bit cell based multiple-time programmable physical unclonable function (PUF) for IoT devices security. In: Proceeding 7th IEEE Electron Devices Technology Manufacturing Conference (EDTM), Seoul, South Korea, pp. 1-3. (2023). https://doi.org/10.1109/EDTM55494.2023.10103114
100. Breyer, E. T., Mulaosmanovic, H., Mikolajick, T., Slesazeck, S.: Reconfigurable NAND/NOR logic gates in 28 nm HKMG and 22 nm FD-SOI FeFET technology. In: Proceeding IEEE International Electron Devices Meeting (IEDM), San Francisco, CA, USA, (2017). https://doi.org/10.1109/IEDM.2017.8268471
101. Breyer, E. T., Mulaosmanovic, H., Slesazeck, S., Mikolajick, T.: Demonstration of versatile nonvolatile logic gates in 28 nm HKMG FeFET technology. In: Proceeding IEEE International Symposium Circuits System (ISCAS), Florence, Italy, pp. 1-5 (2018)
102. Ramanujam, S., Burleson, W.: Reconfiguring the Mux-based Arbiter PUF using FeFETs. In: Proceeding 22nd International Symposium Quality Electronic Design (ISQED), Santa Clara, CA, USA, pp. 257-262 (2021). https://doi.org/10.1109/ISQED51717.2021.9424328
103. Guo, X., Ma, X., Müller, F., Dunkel, S.: Exploiting FeFET switching stochasticity for low-power reconfigurable physical unclonable function. In: Proceeding 47th Europe Solid-State Circuits Conference (ESSCIRC), Grenoble, France, pp. 119-122 (2021)
104. Zhou, Y., Liang, Z., Luo, W., Yu, M.: Ferroelectric and interlayer co-optimization with in-depth analysis for high endurance

FeFET. In: Proceeding IEEE International Electron Devices Meeting (IEDM), San Francisco, CA, USA, (2022). https://doi.org/10.1109/IEDM45625.2022.10019465
105. Shao, H., Zhou, Y., Huang, W., Su, C.: A novel FeFET array-based PUF: co-optimization of entropy source and CRP generation for enhanced robustness in IoT security. In: Proceeding IEEE International Electron Devices Meeting (IEDM), San Francisco, CA, USA, pp. 1-4 (2023). https://doi.org/10.1109/IEDM45741.2023.10413787
106. Xu, Y., Xiao, Y., Zhao, Z., Müller, F., Vardar, A., Gong, X., George, S., Kämpfe, T., Narayanan, V., Ni, K.: Embedding security into ferroelectric FET array via in situ memory operation. Nat. Commun., vol. 14, no. 1, p. 8287, (2023). https://doi.org/10.1038/s41467-023-43941-5
107. Lim, S., Hwang, J., Ko, D.H., Kim, S.K.: Design of physically unclonable function using ferroelectric FET with auto write-back technique for resource-limited IoT security. IEEE Internet Things J. **11**(16), 27676–27686 (2024). https://doi.org/10.1109/JIOT.2024.3399482
108. Li, T., Guo, X., Müller, F., Abdulazhanov, S., Ma, X., Zhong, H., Liu, Y., Narayanan, V., Yang, H., Ni, K., Kämpfe, T., Li, X.: Demonstration of high-reconfigurability and low-power strong physical unclonable function empowered by FeFET cycle-to-cycle variation and charge-domain computing. Nat. Commun. **16**(1), 189 (2025). https://doi.org/10.1038/s41467-024-55380-x
109. Yu, J., Min, K.K., Kim, Y., Kim, S., Hwang, S., Kim, T.H., Kim, C., Kim, H., Lee, J.-H., Kwon, D., Park, B.-G.: A novel physical unclonable function (PUF) using 16 × 16 pure-HfOxferroelectric tunnel junction array for security applications. Nanotechnology **32**(48), 485202 (2021). https://doi.org/10.1088/1361-6528/ac1dd5
110. Kim, S., Lee, K., Oh, M.H., Lee, J.H.: Physical unclonable functions using ferroelectric tunnel junctions. IEEE Electron Device Lett. **42**(6), 816–819 (2021). https://doi.org/10.1109/LED.2021.3075427
111. Lim, S., Goh, Y., Lee, Y. K., Ko, D. H.: A highly integrated cross-point array using self-rectifying FTJ for dual-mode operations: CAM and PUF. In: Proceeding 48th Europe Solid-State Circuits Conference (ESSCIRC), Milan, Italy, pp. 113-116 (2022). https://doi.org/10.1109/ESSCIRC55480.2022.9911355
112. Noor, N., Silva, H.: Phase change memory for physical unclonable functions. in Applications of Emerging Memory Technology: Beyond Storage, Singapore, pp. 59–91 (2019)
113. Cattaneo, L., Ielmini, D.: A strong physical unclonable function with virgin state embedded phase change memory. IEEE Trans. Electron Devices **71**(10), 6032–6039 (2024). https://doi.org/10.1109/TED.2024.3449256
114. Kaur, M., Rashidzadeh, R., Muscedere, R.: Reliability of Physical Unclonable Function under Temperature and Supply Voltage Variations. In: 2018 IEEE 61st International Midwest Symposium on Circuits and Systems (MWSCAS), pp. 1008-1011, (2018). https://doi.org/10.1109/MWSCAS.2018.8623849.
115. Mahmoodi, M.R., Strukov, D.B., Kavehei, O.: Experimental demonstrations of security primitives with nonvolatile memories. IEEE Trans. Electron Devices **66**(12), 5050–5059 (2019). https://doi.org/10.1109/TED.2019.2948950
116. Chen, Z., Wu, M., Zhou, Y., Li, R., Tan, J., Ding, D.: PUF-CIM: SRAM-based compute-in-memory with zero bit-error-rate physical unclonable function for lightweight secure edge computing. IEEE Trans. Very Large Scale Integr. (VLSI) Syst., vol. 31, no. 8, pp. 1234–1247, (2023). https://doi.org/10.1109/TVLSI.2023.3277517
117. Sun, G., Niu, D., Ouyang, J., Xie, Y.: A frequent-value based PRAM memory architecture. 16th Asia and South Pacific Design Automation Conference (ASP-DAC), Yokohama, Japan, pp. 211–216, (2011). https://doi.org/10.1109/ASPDAC.2011.5722186
118. Cho, S., Lee, H.: Flip-N-Write: A simple deterministic technique to improve PRAM write performance, energy and endurance. 2009 42nd Annual IEEE/ACM International Symposium on Microarchitecture (MICRO), New York, NY, USA, pp. 347–357, (2009). https://doi.org/10.1109/MICRO.2009.5461046
119. Kim, K.-H., Karpov, I., Olsson, R.H., III., Jariwala, D.: Wurtzite and fluorite ferroelectric materials for electronic memory. Nat. Nanotechnol. **18**, 422–441 (2023). https://doi.org/10.1038/s41565-023-01361-y
120. Chen, A.: Reconfigurable physical unclonable function based on probabilistic switching of RRAM. Electron. Lett. **51**(8), 615–617 (2015). https://doi.org/10.1049/el.2014.4375
121. Shannon, C.E.: A mathematical theory of communication. Bell Syst. Tech. J. **27**(3), 379–423 (1948). https://doi.org/10.1002/j.1538-7305.1948.tb01338.x
122. Rukhin, A., Soto, J., Nechvatal, J., Smid, M.: A statistical test suite for random and pseudorandom number generators for cryptographic applications, vol. 800-22. NIST Spec. Publ., (2001)
123. Zhang, L., Fong, X., Chang, C.-H., Kong, Z.H., Roy, K.: Highly reliable spin-transfer torque magnetic RAM-based physical unclonable function with multi-response-bits per cell. IEEE Trans. Inf. Forensics Secur. **10**(8), 1630–1642 (2015). https://doi.org/10.1109/TIFS.2015.2421481
124. Wang, D. Y., Hsin, Y. C., Lee, K. Y., Chen, G. L.: Hardware implementation of physically unclonable function (PUF) in perpendicular STT MRAM. In: Proceeding International Symposium VLSI Technology System Applications (VLSI-TSA), Hsinchu, Taiwan, pp. 1-2 (2017). https://doi.org/10.1109/VLSI-TSA.2017.7942497
125. Hou, Z., Wang, Z., Wang, C., Wang, M.: Reconfigurable and dynamically transformable In-Cache-MPUF system with true randomness based on the SOT-MRAM. IEEE Trans. Circuits Syst. I Regul. Pap., vol. 69(7)2694–2706, (2022). https://doi.org/10.1109/TCSI.2022.3168133
126. Zhang, S., Zhang, J., Li, S., Wang, Y.: Reconfigurable physical unclonable cryptographic primitives based on current-induced nanomagnets switching. Sci. China Inf. Sci. **65**(2), 122405 (2022). https://doi.org/10.1007/s11432-021-3270-8
127. John, R.A., Shah, N., Vishwanath, S.K., Ng, S.E., Febriansyah, B., Jagadeeswararao, M., Chang, C.-H., Basu, A., Mathews, N.: Halide perovskite memristors as flexible and reconfigurable physical unclonable functions. Nat. Commun., vol. 12, p. 3681, (2021). https://doi.org/10.1038/s41467-021-24057-0
128. Pham, V. K., Ngo, C. T., Nam, J. -W., Hong, J. -P.: A reconfigurable SRAM CRP PUF with high reliability and randomness, Electronics, vol. 13, no. 2, p. 309, (2024), https://doi.org/10.3390/electronics13020309.
129. Li, J., Cui, Y., Gu, C., Wang, C., Liu, W., Kvatinsky, S.: A highly reliable dual-mode RRAM PUF with key concealment scheme. IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst., vol. 44(8), pp. 2870–2882 (2025). https://doi.org/10.1109/TCAD.2025.3536376
130. Bai, Y., Wu, H., Wu, R., Zhang, Y., Deng, N., Yu, Z., Qian, H.: Study of multi-level characteristics for 3D vertical resistive switching memory. Sci. Rep., vol. 4, p. 5780, (2014). https://doi.org/10.1038/srep05780
131. Rzeszut, P., Chęciński, J., Brzozowski, I., Ziętek, S., Skowroński, W., Stobiecki, T..: Multi-state MRAM cells for hardware neuromorphic computing. Sci. Rep., vol. 12, p. 7178, (2022). https://doi.org/10.1038/s41598-022-11199-4
132. Lee, G. S., Ju, H.: Investigation of entropy space affecting on novel MLC-ReRAM based PUF and system level optimization. Proceeding 4th World Congress on New Technologies, (2018). https://api.semanticscholar.org/CorpusID:116789265

133. Lee, G.S., Kim, G.-H., Kwak, K., Jeong, D.S., Ju, H.: Enhanced reconfigurable physical unclonable function based on stochastic nature of multilevel cell RRAM. IEEE Trans. Electron Devices **66**(4), 1717–1721 (2019). https://doi.org/10.1109/TED.2019.2898455

134. Yue, W., Wu, K., Li, Z., Zhou, J., Wang, Z., Zhang, T., Yang, Y., Ye, L., Wu, Y., Bu, W., Wang, S., He, X., Yan, X., Tao, Y., Yan, B., Huang, R., Yang, Y.: Physical unclonable in-memory computing for simultaneous protecting private data and deep learning models, Nat. Commun., vol. 16, p. 1031, (2025), https://doi.org/10.1038/s41467-025-56412-w.

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